

AEO «Nazarbayev intellectual schools»
Center of pedagogical measurements



**EXAMPLE CANDIDATE
RESPONSE OF EXTERNAL
SUMMATIVE ASSESSMENT**

12 grade

CHEMISTRY
Paper 2

Astana, 2025 year

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INTRODUCTION

This booklet presents a collection of example candidate responses from the External Summative Assessment in Chemistry for Grade 12 students at Nazarbayev Intellectual Schools. The assessment is based on the content and expectations of the international Advanced Level Chemistry curriculum.

The aim of this booklet is to support students and teachers in preparing for future assessments by providing real student responses, examiner comments, and mark allocations. The examples are taken from Papers 2 and 3 and are categorized into three levels of performance: A, C, and E, as described in the Chemistry Test Specification.

Examiners marked student work using the official mark scheme, which includes standard abbreviations such as “ecf” (error carried forward). This is applied when a student makes an error in an earlier part of their response, but subsequent steps remain logically valid. The number in square brackets indicates the maximum mark for each response.

By reviewing the sample answers and examiner feedback in this booklet, users can gain a clearer understanding of the assessment criteria, identify common strengths and errors, and enhance teaching approaches in line with the NIS-Programme — the Educational Programme of Nazarbayev Intellectual Schools AEO for Grades 11 and 12 — and international standards.

Question 1 (a), (b) (i)

- 1 Chemical elements are arranged according to their atomic number in the Periodic Table. In the Periodic Table, the properties of elements are described based on their position, moreover, various trends in properties can be predicted across the periods and down the group.

(a) This question is about the elements in Period 3.

State and explain the general trend in atomic radius of the elements across Period 3.

.....

.....

.....

.....

..... [3]

(b) (i) Arrange the cations Na^+ , Mg^{2+} , Al^{3+} in increasing order of ionic radius.

.....

..... [1]

Mark Scheme

1 (a)	(Atomic radius) shows a general <u>decrease</u> ;	1	Allow same number of shells
	The number of protons increases OR nuclear charge increases/attraction of (outer) electrons to nucleus increases	1	
	Shielding is similar/same OR electrons are added to the same shell.	1	
		[3]	
1 (b) (i)	Al^{3+} , Mg^{2+} , Na^+	[1]	

Example candidate response—grade A

- 1 Chemical elements are arranged according to their atomic number in the Periodic Table. In the Periodic Table, the properties of elements are described based on their position, moreover, various trends in properties can be predicted across the periods and down the group.

For
Examiner's
Use

- (a) This question is about the elements in Period 3.

State and explain the general trend in atomic radius of the elements across Period 3.

Across period 3 the atomic radius of
an elements decrease ✓ this caused by increase
in proton number while shell number remains ✓
the same. Greater proton number means nucleus
more positively charged strongly attracting with
electrons. [3]

- (b) (i) Arrange the cations Na^+ , Mg^{2+} , Al^{3+} in increasing order of ionic radius.

~~$\text{Na}^+ \rightarrow \text{Mg}^{2+} \rightarrow \text{Al}^{3+}$~~
 $\text{Al}^{3+} \rightarrow \text{Mg}^{2+} \rightarrow \text{Na}^+$ ✓ [1]

Examiner's comment

The candidate earns 3 out of 3 marks for this question. 1 mark is awarded for stating the general trend that atomic radius decreases across the third period and for explaining this in terms of increasing nuclear charge with the same number of shells.

In part (b)(i), the candidate correctly arranges the cations in increasing order of ionic radius.

Example candidate response—grade C

- 1 Chemical elements are arranged according to their atomic number in the Periodic Table. In the Periodic Table, the properties of elements are described based on their position, moreover, various trends in properties can be predicted across the periods and down the group.

For
Examiner's
Use

- (a) This question is about the elements in Period 3.

State and explain the general trend in atomic radius of the elements across Period 3.

Atomic radius decreases from left to right side of Period 3, because, more electrons, more shells across the period. Distance between nucleus and outermost electrons increase ~~and~~

[3]

- (b) (i) Arrange the cations Na^+ , Mg^{2+} , Al^{3+} in increasing order of ionic radius.

Na^+ , Mg^{2+} , Al^{3+}

[1]

Examiner's comment

The candidate earns 2 out of 3 marks for this question. They correctly describe the trend in atomic radius and explain it in terms of distance but do not mention the second key point involving the same number of shells or similar shielding.

In part (b)(i), the arrangement of the cations in increasing order of ionic radius is incorrect.

Example candidate response—grade E

- 1 Chemical elements are arranged according to their atomic number in the Periodic Table. In the Periodic Table, the properties of elements are described based on their position, moreover, various trends in properties can be predicted across the periods and down the group.

For
Examiner's
Use

- (a) This question is about the elements in Period 3.

State and explain the general trend in atomic radius of the elements across Period 3.

Atomic radius from Na to Cl increases, since more electrons arrive on new p-subshell. Meaning that Chlorine is one of the most "big" elements across period 3 in terms of atomic radius. Number of electrons increase, new sub-shells occur less attraction to the nucleus from outer shell. [3]

- (b) (i) Arrange the cations Na^+ , Mg^{2+} , Al^{3+} in increasing order of ionic radius.

$\text{Na}^+ \rightarrow \text{Mg}^{2+} \rightarrow \text{Al}^{3+}$

[1]

Examiner's comment

The candidate demonstrates an incorrect understanding of the trend in atomic radius across the third period, failing to link the trend to increasing nuclear attraction within the atom. Additionally, they do not mention that the elements in the third period have the same number of electron shells.

Furthermore, the arrangement of the cations in increasing order of ionic radius is also incorrect.

Question 1 (b) (ii)

- (ii) Explain the difference in size between a magnesium atom and a magnesium ion.

.....

.....

.....

..... [2]

Mark Scheme

1 (b) (ii)	Mg ²⁺ contain fewer electrons than protons	1	OWTTE Ignore comparison
	Nucleus has a greater attraction/ electron is closer to the nucleus	1	
		[2]	

Example candidate response—grade A

- (ii) Explain the difference in size between a magnesium atom and a magnesium ion. ✓

Magnesium ion is smaller because the number of protons is more than electrons, while in magnesium atom it is the same. So, electrons in magnesium ion are experience more effective nuclear charge and, therefore, are closer to nucleus. ✓ [2]

2 2

Examiner's comment

The candidate demonstrates an understanding that the Mg²⁺ ion has more protons than electrons, which causes the remaining electrons to be closer to the nucleus. This precise and thorough explanation earns full marks.

Example candidate response—grade C

- (ii) Explain the difference in size between a magnesium atom and a magnesium ion.

Mg⁰ has greater size than Mg²⁺ because Mg²⁺ ion has slightly positive charge due to decreased electron number X which leads to increased attraction between nucleus and outer shell electrons which decreases ionic radius. [2] 1/1

Examiner's comment

The candidate notes a difference in size resulting from nuclear attraction but does not explain how this relates to the differing numbers of protons and electrons. As a result, they receive 1 out of 2 marks.

Example candidate response—grade E

- (ii) Explain the difference in size between a magnesium atom and a magnesium ion.

Magnesium 2+ ions has donated two electrons, meaning that outer shell now has maximum of 2p⁶ configuration. Meanwhile Mg atom has full 3s² electrons, so its size is bigger. Mg > Mg²⁺ XX [2] 0

Examiner's comment

In their answer, the candidate provides an incomplete explanation and compares Mg and Mg²⁺ without explaining why. They do not describe the difference in terms of the number of protons and electrons, or the resulting attraction to the nucleus.

Question 1 (c) (i),(ii)

(c) Each element has a value for its first ionisation energy. This value is given in the *Data Booklet*.

(i) Define the term *first ionisation energy*.

.....

 [2]

(i) Write an equation to represent the first ionisation energy of phosphorus.

.....
 [1]

Mark Scheme

1 (c) (i)	Enthalpy/energy (needed) to remove one mole of electrons from one mole of gaseous atoms; To form one mole of gaseous 1+ ions / ions with +1 charge	1 1 [2]	
1 (c) (ii)	$P(g) \rightarrow P^+(g) + e^-$ State symbols must be included for the mark	[1]	Allow $P(g) - e^- \rightarrow P^+(g)$

Example candidate response—grade A

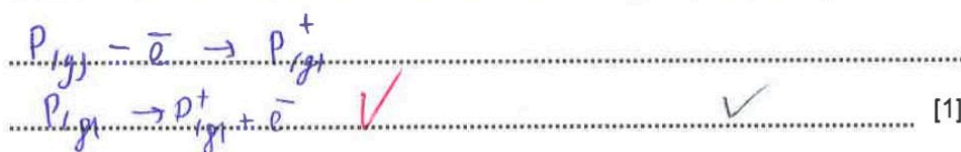
(c) Each element has a value for its first ionisation energy. This value is given in the *Data Booklet*.

(i) Define the term *first ionisation energy*.

enthalpy change when one mole
of electron is removed from one mole ✓
of gaseous atom to form one mole ✓
of +1 charged gaseous ion ✓ [2]

2 2

(ii) Write an equation to represent the first ionisation energy of phosphorus.



For
Examiner's
Use

1 1

Examiner's comment

The candidate includes all key phrases required to define first ionisation energy — “one mole of electrons,” “from one mole of gaseous atoms,” and “to form one mole of positive ions.” They also provide the correct equation for the first ionisation energy of phosphorus.

Example candidate response — grade C

(c) Each element has a value for its first ionisation energy. This value is given in the *Data Booklet*.

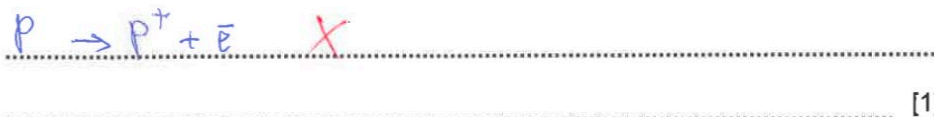
(i) Define the term *first ionisation energy*.

Amount of energy required to remove one
mole of electron from one mole of gaseous
atom in gaseous state X

[2]

1

(ii) Write an equation to represent the first ionisation energy of phosphorus.



[1]

For
Examiner's
Use

0

Examiner's comment

The candidate includes only some key phrases in the definition — “one mole of electrons” and “one mole of gaseous atoms” — but omits the essential phrase “to form one mole of gaseous positive ions,” which is required for a complete answer.

The equation provided for the first ionisation energy of phosphorus does not include state symbols.

Example candidate response—grade E

(c) Each element has a value for its first ionisation energy. This value is given in the *Data Booklet*.

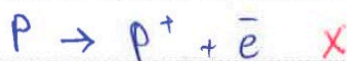
(i) Define the term *first ionisation energy*.

First ionisation energy is the energy required for to take one electron from outer shell of stable elements orbit. xx

[2]

0

(ii) Write an equation to represent the first ionisation energy of phosphorus.



For
Examiner's
Use

[1]

0

Examiner's comment

The candidate demonstrates only a general understanding of the first ionisation energy, without including some of the key phrases "one mole of gaseous atoms," "from one mole of gaseous atoms," and "to form one mole of positive ions."

The equation provided for the first ionisation energy of phosphorus does not include state symbols.

Question 1 (c) (iii), (iv)

(i) Fig. 1 represents the first ionisation energy values of elements across Period 3.

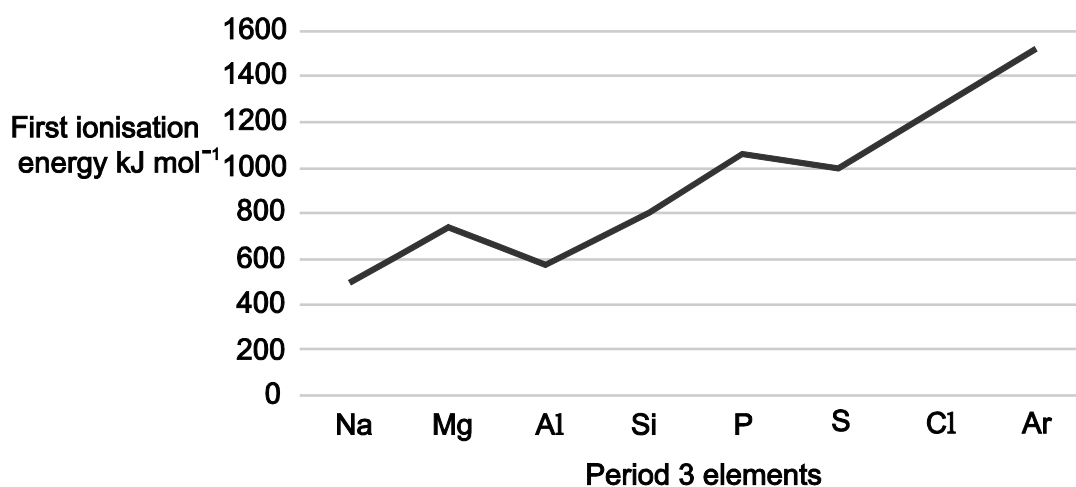


Fig. 1

Draw a cross (x) on Fig. 1 to show the second ionisation energy of phosphorus.

[1]

- (ii) Explain the difference between the first and second ionisation energies of phosphorus.

[1]

Mark Scheme

1 (c) (iii)	Cross above 1012	[1]	
1 (c) (iv)	Electrons are now being removed from positive ions/effective nuclear charge	[1]	OWTTE

Example candidate response—grade A

- (iii) Fig. 1 represents the first ionisation energy values of elements across Period 3.

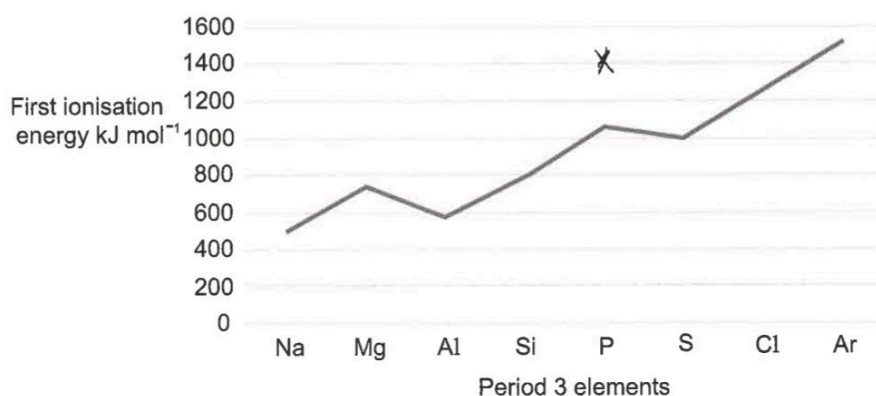


Fig. 1

Draw a cross (x) on Fig. 1 to show the second ionisation energy of phosphorus.

V ✓ [1]

1 1

- (iv) Explain the difference between the first and second ionisation energies of phosphorus.

"first" is removing from gas one mole;
 "second" is removing from (+1 charged) ✓ [1]
 gaseous ion one mole of electron again.

[Total: 11]

1 1
 9
 10

Examiner's comment

The answer correctly shows the second ionisation energy of phosphorus on the graph. A good explanation is given for why the second ionisation energy is higher, noting that the electron is removed from a positive ion.

Example candidate response—grade C

- (iii) Fig. 1 represents the first ionisation energy values of elements across Period 3.

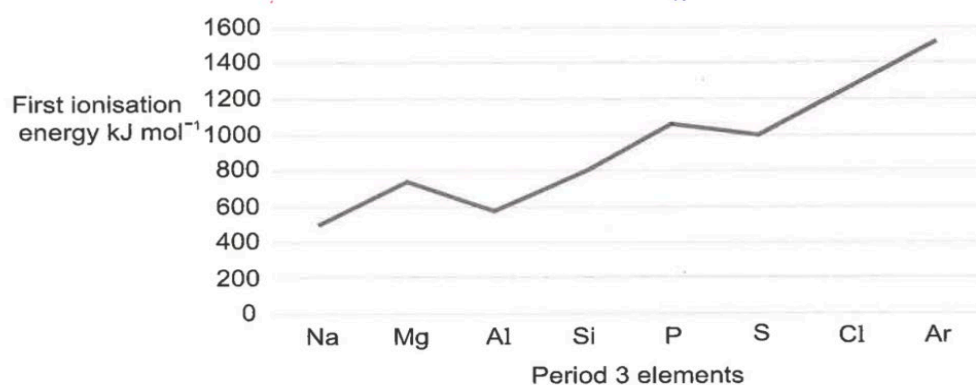
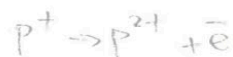


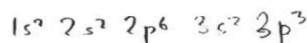
Fig. 1

Draw a cross (x) on Fig. 1 to show the second ionisation energy of phosphorus.



15

[1]



- (iv) Explain the difference between the first and second ionisation energies of phosphorus.

Second ionisation energy is higher because P more energy is needed to extract electron from ^{first} sub-shell. [1]

[Total: 11]

Examiner's comment

The candidate correctly shows the second ionisation energy of phosphorus on the graph. However, they do not explain the difference between the first and second ionisation energies in terms of electrons being removed from positive ions.

Example candidate response—grade E

- (iii) Fig. 1 represents the first ionisation energy values of elements across Period 3.

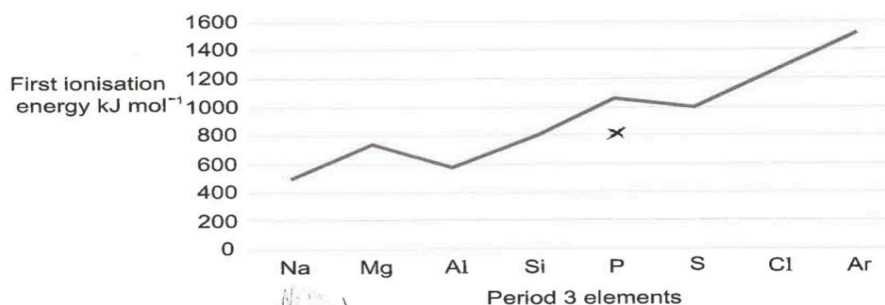


Fig. 1

Draw a cross (x) on Fig. 1 to show the second ionisation energy of phosphorus.

x [1]

- (iv) Explain the difference between the first and second ionisation energies of phosphorus.

On the first ionisation energy of phosphorus, it is a lot smaller than the second ionisation energy of phosphorus. In the first ionisation energy it is 1060 kJ mol⁻¹, and in the second one, it is 1900 kJ mol⁻¹.

[Total: 11]

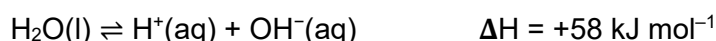
Examiner's comment

The candidate incorrectly shows the second ionisation energy as lower than the first on the graph.

In the following question, they state that the second ionisation energy is higher but do not explain that this is because electrons are being removed from positive ions.

Question 2 (a)

The dissociation of water is a reversible reaction.



- (a) State **two** characteristics of a reversible reaction at equilibrium.

1

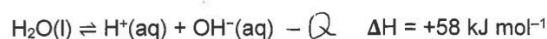
2

Mark Scheme

2(a)	✓rate of forward reaction = rate reverse reaction	1	
	✓macroscopic properties (P,V,T) remain constant/ concentrations remain constant	1 [2]	

Example candidate response—grade A

- 2 The dissociation of water is a reversible reaction.



- (a) State **two** characteristics of a reversible reaction at equilibrium.

- 1 Forward and backward reaction rates are equal to each other ✓
- 2 Concentration of reagent and concentration of products are constant ✓

[2]

For
Examiner's
Use

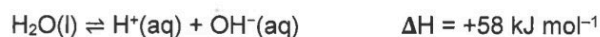
2

Examiner's comment

The candidate clearly shows that in a reversible reaction at equilibrium, the rates of the forward and reverse reactions are equal (1 mark), and the concentrations of the reactants and products remain constant (2 marks).

Example candidate response—grade C

- 2 The dissociation of water is a reversible reaction.



- (a) State **two** characteristics of a reversible reaction at equilibrium.

- 1 forward reaction = backward reaction ✗
- 2 Concentration of products and reactants doesn't change ✓

[2]

For
Examiner's
Use

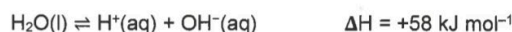
1

Examiner's comment

The candidate does not use the key word "rate" and therefore loses 1 mark. However, they state that the concentrations of the reactants and products do not change and thus gain the second mark.

Example candidate response—grade E

- 2 The dissociation of water is a reversible reaction.



For
Examiner's
Use

- (a) State **two** characteristics of a reversible reaction at equilibrium.

- 1 reaction is endothermic X
- 2 reaction is spontaneous X

[2]

Examiner's comment

The candidate provides some general characteristics of the reaction but does not refer to a reversible reaction at equilibrium and therefore loses both marks.

Question 2(b)

- (b) (i) Write an expression for the ionic product of water, K_w .

Include the unit for the ionic product of water.

[2]

- (ii) State and explain how an increase in temperature affects the value of the ionic product of water, K_w .

.....

.....

.....

..... [2]

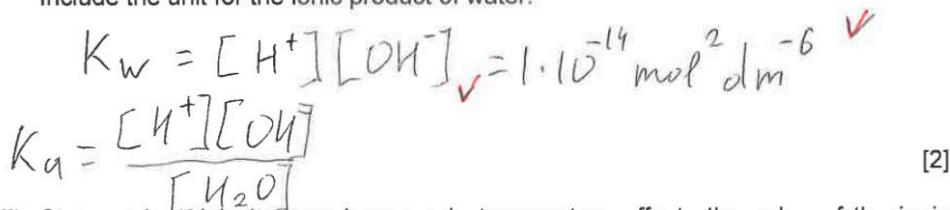
Mark Scheme

2(b)(i)	$K_w = [\text{H}^+][\text{OH}^-]$ $\text{mol}^2 \text{ dm}^{-6}$	1 1 [2]	
2(b)(ii)	K_w will be higher in high temperature/ increase because reaction is endothermic/more ion H^+ , OH^- / equilibrium shift to the right	1 1 [2]	

Example candidate response—grade A

- (b) (i) Write an expression for the ionic product of water, K_w .

Include the unit for the ionic product of water.



- (ii) State and explain how an increase in temperature affects the value of the ionic product of water, K_w .

By Le Chatelier's principle, reaction will shift at right, concentrations of H^+ and OH^- will increase so K_w will increase as well. (Shift towards endothermic)

[2]

Examiner's comment

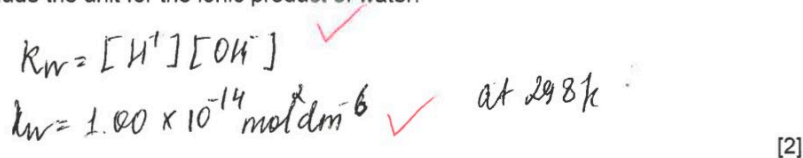
The candidate writes the correct expression for the ionic product of water (1 mark), including the correct units (2 marks).

They also correctly state and explain how an increase in temperature affects the value of K_w , referring to Le Chatelier's principle and noting that the increased concentration of ions (1 mark) results in an increase in K_w (2 mark).

Example candidate response—grade C

- (b) (i) Write an expression for the ionic product of water, K_w .

Include the unit for the ionic product of water.



- (ii) State and explain how an increase in temperature affects the value of the ionic product of water, K_w .

K_w will decrease. Increase

[2]

Examiner's comment

The candidate correctly writes the expression for the ionic product of water with the correct unit and gains 2 marks.

They state the effect of temperature on K_w , but do not explain it in terms of increasing ion concentrations and therefore gain only 1 mark.

Example candidate response—grade E

- (b) (i) Write an expression for the ionic product of water, K_w .

Include the unit for the ionic product of water.

$$K_w = \frac{[H^+][OH^-]}{[H_2O]}$$

[2]

- (ii) State and explain how an increase in temperature affects the value of the ionic product of water, K_w .

The value of the ionic product of water will
increase ✓

[2]

Examiner's comment

The candidate writes an incorrect expression for K and does not include the unit, so no marks are awarded.

They state that the ionic product increases with temperature but do not provide an explanation and therefore gain only 1 mark.

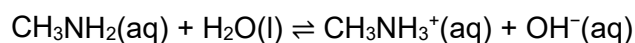
Question 2(c) (i), (iv)

- (c) K_w is also used to calculate the pH of various solutions.

- (i) Use the value of K_w from the *Data Booklet* to calculate the pH of 0.010 mol dm⁻³ KOH.

pH = [2]

- (i) Methylamine is a weak base, which dissociates partially in water.



Write down the expression for K_b for this reaction.

[1]

- (ii) K_b for this reaction is $4.38 \times 10^{-4} \text{ mol dm}^{-3}$. Calculate the pH of $0.010 \text{ mol dm}^{-3} \text{CH}_3\text{NH}_2$.

pH =[4]

- (iii) Explain the difference between pH values calculated in **c (i)** and **c (iii)**.

.....
 [1]

Mark Scheme

2(c)(i)	$[\text{H}^+] = (1 \times 10^{-14}) / 1 \times 10^{-2} = 1 \times 10^{-12}$ $\text{pH} = -\log_{10}[\text{H}^+] = 12$	1 1 [2]	ecf correct ans = [2]
2(c)(ii)	$[\text{CH}_3\text{NH}_3^+][\text{OH}^-] / [\text{CH}_3\text{NH}_2]$	[1]	Do not allow $[\text{CH}_3\text{NH}_3^+][\text{OH}^-] / [\text{CH}_3\text{NH}_2][\text{H}_2\text{O}]$
2(c)(iii)	$[\text{OH}^-] = \sqrt{4.38} \times 10^{-4} \times 0.010$ $[\text{OH}^-] = 2.09 \times 10^{-3}$ $[\text{H}^+] = 4.78 \times 10^{-12}$ $\text{pH} = 11.3$	1 1 1 1 [4]	Allow M1 for correct calculation of $[\text{OH}^-]$ Ecf correct answer [4] Do not accept 11
2(c)(iv)	Methylamine is partially dissociated/ionised OR KOH is completely dissociated/ionised OR KOH is strong base	[1]	OWTTE Do not accept methylamine is weak base

Example candidate response—grade A

(c) K_w is also used to calculate the pH of various solutions.

- (i) Use the value of K_w from the *Data Booklet* to calculate the pH of $0.010 \text{ mol dm}^{-3}$ KOH.

$$[H^+] = \frac{K_w}{[OH^-]} \quad [OH^-] = 0.010 \text{ mol dm}^{-3}$$

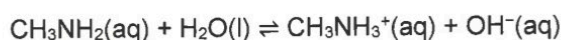
$$[H^+] = 10^{-12} \text{ mol dm}^{-3}$$

$$pH = -\log [H^+] = 12$$

pH =12.0 [2]

2

- (ii) Methylamine is a weak base, which dissociates partially in water.



Write down the expression for K_b for this reaction.

$$K_b = \frac{[CH_3NH_3^+][OH^-]}{[CH_3NH_2]} = \frac{[OH^-]^2}{[CH_3NH_2]}$$

[1]

For
Examiner's
Use

1

- (iii) K_b for this reaction is $4.38 \times 10^{-4} \text{ mol dm}^{-3}$. Calculate the pH of $0.010 \text{ mol dm}^{-3} CH_3NH_2$.

$$[OH^-] = \sqrt{4.38 \cdot 10^{-4} \cdot 0.01} = 0.00209 \text{ mol dm}^{-3}$$

$$pOH = -\log [OH^-] = 2.68$$

$$pH = 14 - pOH = 11.32$$

pH =11.32 [4]

4

- (iv) Explain the difference between pH values calculated in c (i) and c (iii).

Weak base does not dissociate completely,
therefore OH^- produced is less.

[1]

1

Examiner's comment

The candidate correctly uses the K_w value to calculate the pH of KOH and gains 2 marks.

The expression for K_b is written correctly, earning 1 mark.

The calculation of pH for methylamine is accurate: $[OH^-]$ is correctly determined (2 marks), followed by correct pOH (1 mark) and pH (1 mark).

In part (iv), the candidate gives a correct explanation for the difference in pH values, stating that the weak base does not dissociate completely, resulting in less OH^- being produced — 1 mark awarded.

Example candidate response—grade C

(c) K_w is also used to calculate the pH of various solutions.

(i) Use the value of K_w from the *Data Booklet* to calculate the pH of 0.010 mol dm⁻³ KOH.

$$pH = -\log$$

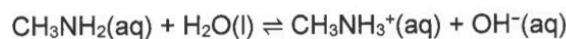
$$K_w = [H^+][OH^-]$$

$$c(OH^-) = 0.010 \text{ mol dm}^{-3}$$

$$[H^+] = \frac{K_w}{[OH^-]} = \frac{1 \cdot 10^{-14}}{0.01} = 1 \cdot 10^{-12} \text{ mol dm}^{-3}$$

$$pH = 12 \quad [2]$$

(ii) Methylamine is a weak base, which dissociates partially in water.



Write down the expression for K_b for this reaction.

$$K_b = \frac{[CH_3NH_3^+][OH^-]}{[CH_3NH_2] + [H_2O]}$$

[1]

(iii) K_b for this reaction is $4.38 \times 10^{-4} \text{ mol dm}^{-3}$. Calculate the pH of 0.010 mol dm⁻³ CH_3NH_2 .

$$4.38 \cdot 10^{-4} = \frac{[OH^-]^2}{0.010}$$

$$[OH^-]^2 = 4.38 \cdot 10^{-6}$$

$$[OH^-] = 2.093 \cdot 10^{-3}$$

$$[OH^-] \quad pH = -\log [2.093 \cdot 10^{-3}] = 2.68$$

$$(H) pH = 14 - 2.68 = 11.32$$

$$pH = 11.32 \quad [4]$$

(iv) Explain the difference between pH values calculated in c (i) and c (iii).

pH in c(i) is more acidic than pH in c(iii). In c(iii) is base.

[1]

For
Examiner's
Use

Examiner's comment

The candidate correctly calculates $[H^+]$ for the strong base but does not convert it to pH, therefore, only 1 mark is awarded.

The expression for the weak base is incorrect — a plus sign is used instead of multiplication, and water is incorrectly included in the expression.

The candidate correctly calculates the concentration of OH^- ions and determines the pH of the weak base, earning full marks for the calculation.

In the comparison, the candidate uses general terms like "more acidic" or "more basic," but does not refer to key reasoning related to dissociation and therefore does not earn the explanation mark.

Example candidate response—grade E

(c) K_w is also used to calculate the pH of various solutions.

- (i) Use the value of K_w from the *Data Booklet* to calculate the pH of $0.010 \text{ mol dm}^{-3}$ KOH.

$$K_w = \frac{[\text{OH}^-]^2}{[\text{KOH}]} \quad 1.00 \cdot 10^{-14} = \frac{[\text{OH}^-]^2}{0.010} \quad [\text{OH}^-]^2 = 1 \cdot 10^{-16}$$

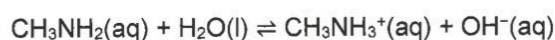
$$[\text{OH}^-] = \sqrt{1 \cdot 10^{-16}} = 1 \cdot 10^{-8}$$

$$\text{pH} = \dots 6 \dots [2]$$

$$\text{pOH} = -\log(1 \cdot 10^{-8}) = 8$$

$$\text{pH} = 14 - 8 = 6$$

- (ii) Methylamine is a weak base, which dissociates partially in water.



Write down the expression for K_b for this reaction.

$$K_b = \frac{[\text{CH}_3\text{NH}_3^+][\text{OH}^-]}{[\text{CH}_3\text{NH}_2][\text{H}_2\text{O}]}$$

[1]

- (iii) K_b for this reaction is $4.38 \times 10^{-4} \text{ mol dm}^{-3}$. Calculate the pH of $0.010 \text{ mol dm}^{-3}$ CH_3NH_2 .

$$\text{pH} = \text{p}K_b + \dots$$

$$\text{pH} = 3.36 + 0.01$$

$$\text{pH} = 3.37$$

$$\text{pH} = \dots 3.37 \dots [4]$$

- (iv) Explain the difference between pH values calculated in c (i) and c (iii).

CH_3NH_2 is weak base which has low pH.

KOH is strong acid

[1]

For
Examiner's
Use

0

0

0

0

Examiner's comment

The candidate uses an incorrect method to calculate the pH of the strong base, resulting in no marks awarded for this part.

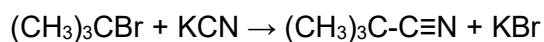
The expression for the weak base is incorrect.

The candidate incorrectly uses $\text{p}K_b$ in the calculation of the pH for the weak base and loses all marks for the calculation.

When comparing, the candidate simply restates that methylamine is a weak base (already stated in the question) and incorrectly describes KOH as a strong acid, therefore, no mark is awarded.

Question 3 (a), (b)

In ethanolic solution, 2-bromo-2-methylpropane, $(\text{CH}_3)_3\text{CBr}$, reacts with potassium cyanide, KCN. This is a nucleophilic substitution reaction.



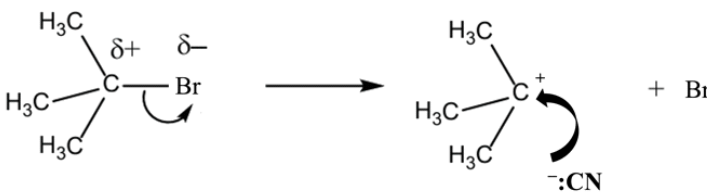
(a) State what is meant by the term *nucleophile*.

..... [1]

(b) Show the mechanism for this reaction. Include all necessary curly arrows, lone pairs, and relevant dipoles.

[3]

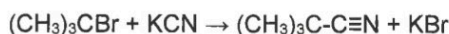
Mark Scheme

3(a)	reagent that forms bonds by donating an electron pair	[1]	Accept lone pair
3(b)	 M1: dipole on C–Br bond AND curly arrow breaking C–Br bond M2: tertiary carbocation and Br[–] (ion) formed in the mechanism M3: arrow from the lone pair of electrons on the carbon atom of the negatively charged cyanide ion to the central C⁺ atom.	1 1 1 [3]	Max 1 mark for the wrong reactant or “sticks” S_N2 – 1 mark

Example candidate response—grade A

- 3 In ethanolic solution, 2-bromo-2-methylpropane, $(\text{CH}_3)_3\text{CBr}$, reacts with potassium cyanide, KCN. This is a nucleophilic substitution reaction.

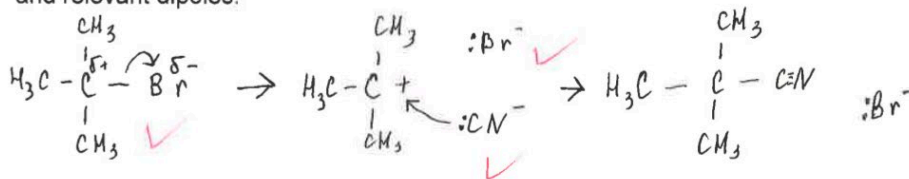
For
Examiner's
Use



- (a) State what is meant by the term *nucleophile*.

nucleophile is a 'nucleo loving' species that are electron rich and can donate a pair of electrons to the electron deficient species. Usually negatively charged. [1]

- (b) Show the mechanism for this reaction. Include all necessary curly arrows, lone pairs, and relevant dipoles.



[3]

3

Examiner's comment

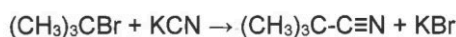
The candidate states that a nucleophile is an electron pair donor.

The dipole on the C–Br bond is correctly shown, along with the curly arrow breaking the C–Br bond, the formation of the tertiary carbocation and Br^- ion, and the arrow from the lone pair of electrons on the carbon atom of the negatively charged cyanide ion to the central C^+ atom.

Example candidate response—grade C

- 3 In ethanolic solution, 2-bromo-2-methylpropane, $(\text{CH}_3)_3\text{CBr}$, reacts with potassium cyanide, KCN. This is a nucleophilic substitution reaction.

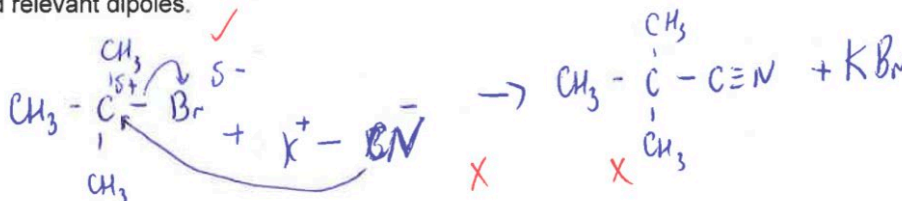
For
Examiner's
Use



- (a) State what is meant by the term *nucleophile*.

molecule that donate pair of electrons [1]

- (b) Show the mechanism for this reaction. Include all necessary curly arrows, lone pairs, and relevant dipoles.



[3]

1

Examiner's comment

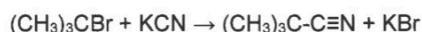
The candidate correctly explains the meaning of a nucleophile.

The dipole on the C–Br bond and the curly arrow breaking the C–Br bond are correctly shown. However, the tertiary carbocation and Br[−] ion are not included in the mechanism, and the arrow from the carbon atom of the negatively charged cyanide ion to the central C⁺ atom is drawn incorrectly.

Example candidate response—grade E

- 3 In ethanolic solution, 2-bromo-2-methylpropane, (CH₃)₃CBr, reacts with potassium cyanide, KCN. This is a nucleophilic substitution reaction.

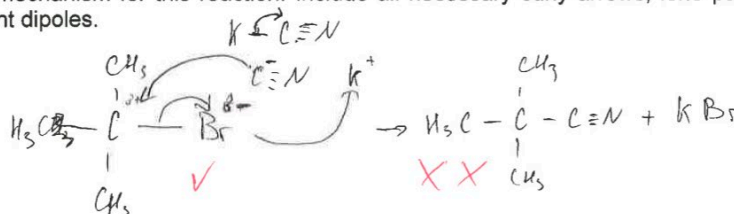
For
Examiner's
Use



- (a) State what is meant by the term *nucleophile*.

particle that have free ~~free~~ pair of e[−]
that can form bonds. X [1]

- (b) Show the mechanism for this reaction. Include all necessary curly arrows, lone pairs, and relevant dipoles.



[3]

Examiner's comment

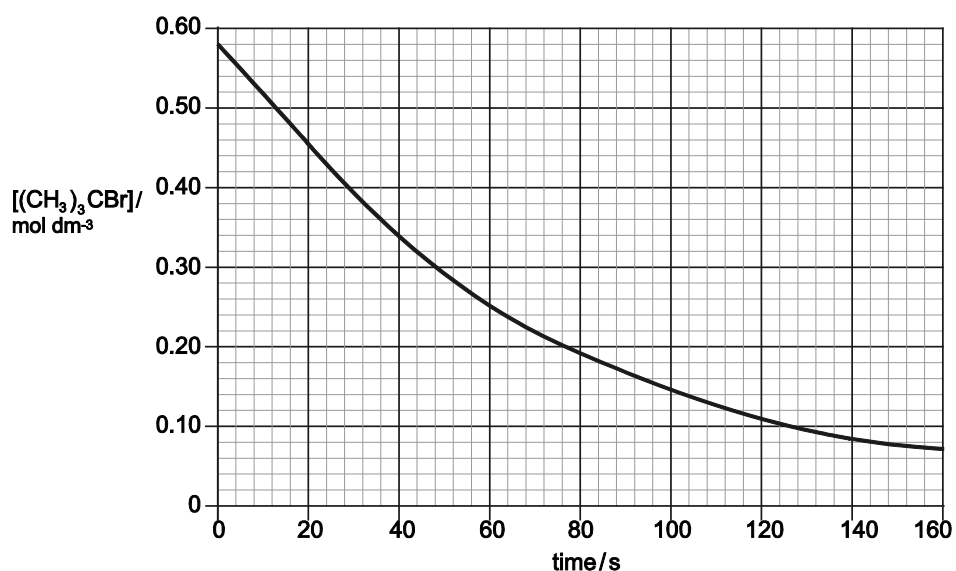
The candidate mentions the electron pair but does not refer to donation.

In the mechanism, the arrow from the C–Br bond to Br[−] is shown with a dipole, but the candidate provides only a general mechanism without the transition stage.

The CN[−] ion is drawn without a lone pair on the carbon, and an incorrect arrow is shown from the bromine atom to H⁺.

Question 3 (c) (i), (ii)

- (c) The kinetics of this reaction is studied and the graph of the concentration of a sample of 2-bromo-2-methylpropane plotted against time is shown below. The rate of this reaction is analysed using potassium cyanide in a large excess.



- (i) Draw a tangent to the curve when the concentration of 2-bromo-2-methylpropane is 0.30 mol dm^{-3} .

[1]

- (ii) The initial rate of the reaction in this experiment has the value $8.81 \times 10^{-3} \text{ mol dm}^{-3} \text{ s}^{-1}$. Use this value and a rate obtained from your tangent, to show that the order of the reaction is 1st with respect to 2-bromo-2-methylpropane.

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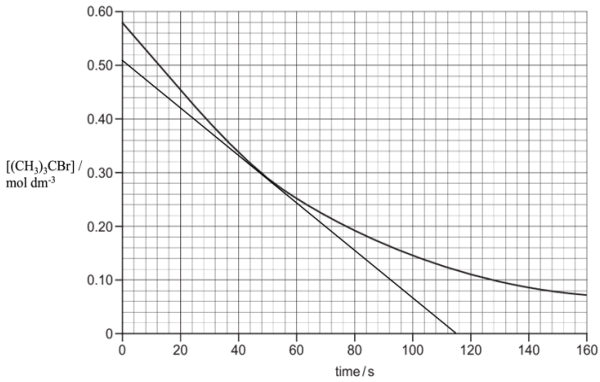
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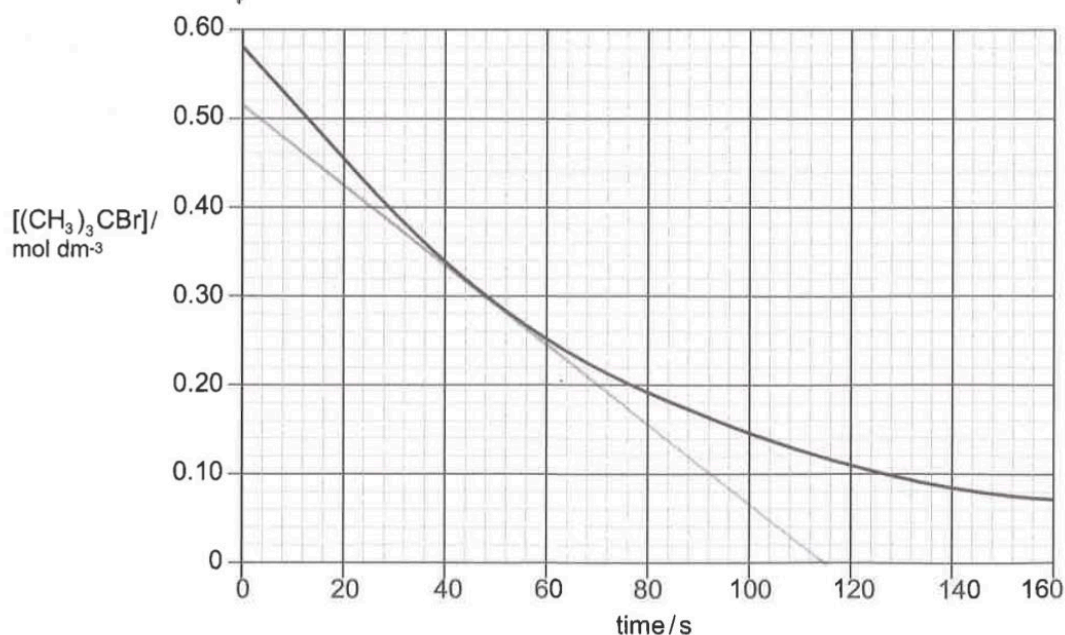
[4]

Mark Scheme

3(c)(i)		[1]	Accept any appropriate tangent. Line must touch the curve at 0.30 but must not cross the curve.
3(c)(ii)	Stage 1: Rate at concentration = 0.30 mol dm^{-3} from the tangent is $0.52 - 0 / 115 - 0 = 4.52 \times 10^{-3}$ Stage 2: Comparison of rates and concentrations Initial rate / rate at 0.30 = $(8.81 \times 10^{-3}) / (4.52 \times 10^{-3}) = 1.95$ Initial concentration / concentration at point where tangent drawn = $0.58 / 0.30 = 1.93$ Stage 3: Deduction of order If order is 1, rate should increase by factor of $(1.93)^1 = 1.93$ this is approximately equal to 1.95 therefore order is 1st with respect to 2-bromo-2-methylpropane	1 1 1 1 [4]	Range between 4.30×10^{-3} and 4.80×10^{-3} Ecf from stage 1 Marking points in stage 2 can be in either order

Example candidate response—grade A

- (c) The kinetics of this reaction is studied and the graph of the concentration of a sample of 2-bromo-2-methylpropane plotted against time is shown below. The rate of this reaction is analysed using potassium cyanide in a large excess.



- (i) Draw a tangent to the curve when the concentration of 2-bromo-2-methylpropane is 0.30 mol dm^{-3} .

For
Examiner
Use

[1]

- (ii) The initial rate of the reaction in this experiment has the value $8.81 \times 10^{-3} \text{ mol dm}^{-3} \text{ s}^{-1}$. Use this value and a rate obtained from your tangent, to show that the order of the reaction is 1st with respect to 2-bromo-2-methylpropane.

$$\text{grad} = \frac{0.52 - 0}{80} = 6.5 \times 10^{-3} \text{ mol dm}^{-3} \text{ s}^{-1} \rightarrow \frac{8.81 \times 10^{-3}}{6.5 \times 10^{-3}} \approx 1.37 \approx 1$$

So the rate of reaction decreased two times at $[(\text{CH}_3)_3\text{CBr}]$ of 0.3 mol dm^{-3} . This means that decreasing the concentration for two times will decrease the rate for two times, meaning the order w.r.t $[(\text{CH}_3)_3\text{CBr}]$ is equal to 1.

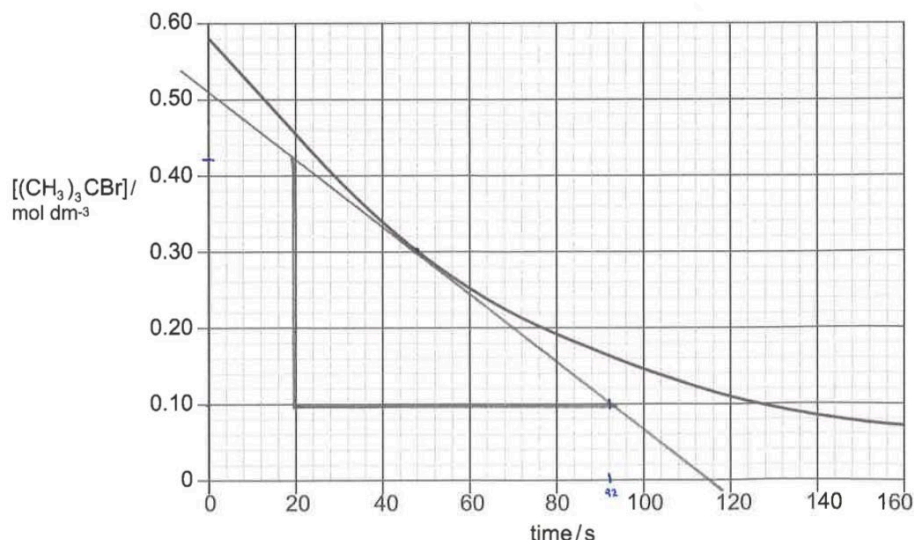
[4]

Examiner's comment

The candidate correctly draws the tangent and calculates the actual rate based on it. They compare the rates using calculations, make a valid comparison of concentrations, and correctly deduce the order of reaction based on the calculated values.

Example candidate response—grade C

- (c) The kinetics of this reaction is studied and the graph of the concentration of a sample of 2-bromo-2-methylpropane plotted against time is shown below. The rate of this reaction is analysed using potassium cyanide in a large excess.



- (i) Draw a tangent to the curve when the concentration of 2-bromo-2-methylpropane is 0.30 mol dm^{-3} .

Exe

✓ [1]

1

- (ii) The initial rate of the reaction in this experiment has the value $8.81 \times 10^{-3} \text{ mol dm}^{-3} \text{ s}^{-1}$. Use this value and a rate obtained from your tangent, to show that the order of the reaction is 1st with respect to 2-bromo-2-methylpropane.

rate = $k[(\text{CH}_3)_3\text{CBr}][\text{KCN}]$. Tangent to the curve helps to measure gradient (slope) of $(\text{CH}_3)_3\text{CBr}$ concentration. $\Delta G = \frac{(0.42 - 0.10) \text{ mol}}{(82 - 20) \text{ s} \cdot \text{dm}^3} = \frac{1}{225} = 0.0044$ ~~$\times 10^{-3}$~~ ~~KCN~~ ~~B, B7~~

$8.81 \cdot 10^{-3} = 0.0044 \cdot k \cdot 0.3 \cdot [\text{KCN}] \Rightarrow \text{KCN} = \text{B, B7}$

So because tangent at 0.3 mol/dm^3 is a straight line and slope is 0.0044 and $0.3/8.81 \cdot 10^{-3} = 0.034$. $0.044 : 0.034$ ratio ~~$\times \times \times$~~

is approximately 1:1, so reaction is 1st order. [4]

1

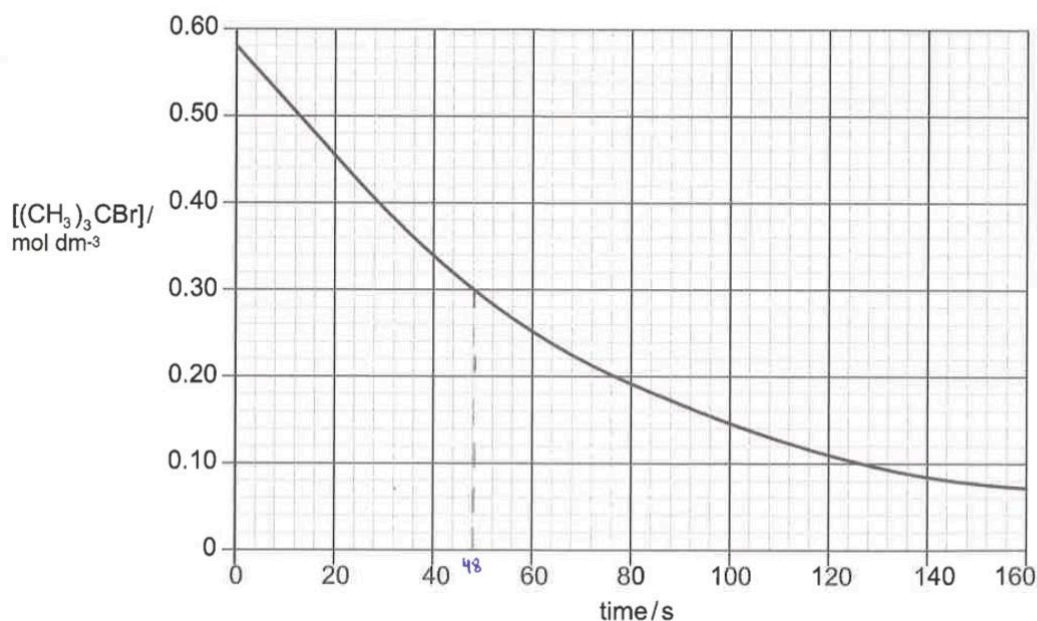
Examiner's comment

The candidate draws a tangent to the curve at 0.3 mol dm^{-3} and correctly calculates the rate at this concentration.

However, they incorrectly use the concentration at the point of the tangent as the initial concentration, leading to an incorrect rate ratio calculation. As a result, the candidate loses 3 marks.

Example candidate response—grade E

- (c) The kinetics of this reaction is studied and the graph of the concentration of a sample of 2-bromo-2-methylpropane plotted against time is shown below. The rate of this reaction is analysed using potassium cyanide in a large excess.



- (i) Draw a tangent to the curve when the concentration of 2-bromo-2-methylpropane is 0.30 mol dm^{-3} .

For
Examiner's
Use

- (ii) The initial rate of the reaction in this experiment has the value $8.81 \times 10^{-3} \text{ mol dm}^{-3} \text{ s}^{-1}$. Use this value and a rate obtained from your tangent, to show that the order of the reaction is 1st with respect to 2-bromo-2-methylpropane.

$$\text{rate} = \frac{C}{t} = \frac{0.30}{48} = 6.25 \cdot 10^{-3} \text{ mol dm}^{-3} \text{ s}^{-1}$$

$$\frac{C_1}{C_2} = \frac{0.58}{0.30} = 1.93$$

$$\frac{\text{rate}_1}{\text{rate}_2} = \frac{8.81 \cdot 10^{-3}}{6.25 \cdot 10^{-3}} =$$

[4]

Examiner's comment

The candidate does not draw a tangent and does not calculate the rate from it.

They compare concentrations but do not complete the comparison of rates, resulting in an incomplete response.

Question 3 (c) (iii), (iv), (d)

- (i) Write the rate equation for the reaction if the order of the reaction with respect to KCN is zero.

[1]

- (ii) Calculate the value for the rate constant, k, and deduce its units.

k = units [1]

- (d) In another experiment 2-iodo-2-methylpropane is used instead of 2-bromo-2-methylpropane. 2-iodo-2-methylpropane reacts faster with KCN than 2-bromo-2-methylpropane. Explain the reason.

.....

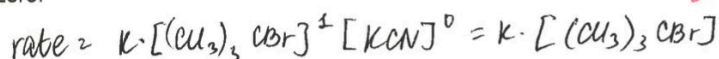
 [2]

Mark Scheme

3(c)(iii)	Rate = k [(CH ₃) ₃ CBr]	[1]	
3(c)(iv)	Rate calculated from tangent/ 0.30 OR initial rate / 0.58 = 15,19 * 10 ⁻³ s ⁻¹ AND s ⁻¹	[1]	Ecf
3 (d)	C-I bond energy is lower than C-Br / C-I bond is weaker than C-Br/ easy to break Iodine less electronegative/large atomic size/more positive/less polar/low electron density	1 1 [2]	

Example candidate response—grade A

- (iii) Write the rate equation for the reaction if the order of the reaction with respect to KCN is zero.



[1]

- (iv) Calculate the value for the rate constant, k , and deduce its units.

$$k = \frac{8.81 \cdot 10^{-3}}{0.58} = 0.015 \quad \frac{\text{mol dm}^{-3} \text{s}^{-1}}{\text{mol dm}^{-3}} = \text{s}^{-1}$$

$$k = 0.015 \text{ units } \text{s}^{-1} \quad [1]$$

- (d) In another experiment 2-iodo-2-methylpropane is used instead of 2-bromo-2-methylpropane. 2-iodo-2-methylpropane reacts faster with KCN than 2-bromo-2-methylpropane. Explain the reason.

C-I is a weaker bond than C-Br, so it is easily broken at small amount of time. C-I bond is weak because iodine radius is bigger than Cl, so weaker force of attraction between electron pair and nucleus.

[Total: 13]

Examiner's comment

The candidate correctly writes the rate equation, recognising that the reaction is zero order with respect to KCN. The rate constant k is correctly calculated, and the unit is correctly identified as s^{-1} .

The candidate explains the faster rate of reaction of 2-iodo-2-methylpropane with KCN by correctly referring to the larger atomic radius of iodine and correctly applying the key idea that the C-I bond is weaker than the C-Br bond.

Example candidate response—grade C

- (iii) Write the rate equation for the reaction if the order of the reaction with respect to KCN is zero.



[1]

- (iv) Calculate the value for the rate constant, k , and deduce its units.

$$k = \frac{8.81 \cdot 10^{-3} \text{ mol dm}^{-3} \text{s}^{-1}}{0.58 \text{ mol dm}^{-3}} = 0.015 \text{ s}^{-1}$$

$$k = 0.015 \text{ units } \text{s}^{-1} \quad [1]$$

- In another experiment 2-iodo-2-methylpropane is used instead of 2-bromo-2-methylpropane. 2-iodo-2-methylpropane reacts faster with KCN than 2-bromo-2-methylpropane. Explain the reason.

C-I bond energy is lower than C-Br bond energy. This means that lower energy amount is required to break a C-I bond thus the reaction will be faster.

[2]

Examiner's comment

The candidate correctly writes the rate equation and calculates the rate constant k with the correct unit.

In part (d), the candidate states that the C–I bond has lower energy but does not explain the reason using key characteristics such as atomic radius or electronegativity. As a result, only 1 mark is awarded for the explanation.

Example candidate response—grade E

- (iii) Write the rate equation for the reaction if the order of the reaction with respect to KCN is zero.



[1]

- (iv) Calculate the value for the rate constant, k , and deduce its units.

$$8.81 \cdot 10^{-3} = k \cdot 0.30$$
$$k = \frac{8.81 \cdot 10^{-3}}{0.30} = \frac{\text{mol dm}^{-3} \text{s}^{-1}}{\text{mol dm}^{-3}} =$$
$$k = 0.098 \text{ units } \text{s}^{-1} \quad [1]$$

- d) In another experiment 2-iodo-2-methylpropane is used instead of 2-bromo-2-methylpropane. 2-iodo-2-methylpropane reacts faster with KCN than 2-bromo-2-methylpropane. Explain the reason.

Because iodine is more reactive than bromine.

~~It can see~~

[2]

Examiner's comment

The candidate correctly writes the rate equation. However, in the calculation of the rate constant, they incorrectly use the initial rate instead of the initial concentration, resulting in an incorrect value for k .

In the explanation of reactivity, the candidate states a property of iodine but does not link it to bond strength or refer to key terms such as atomic radius or electronegativity.

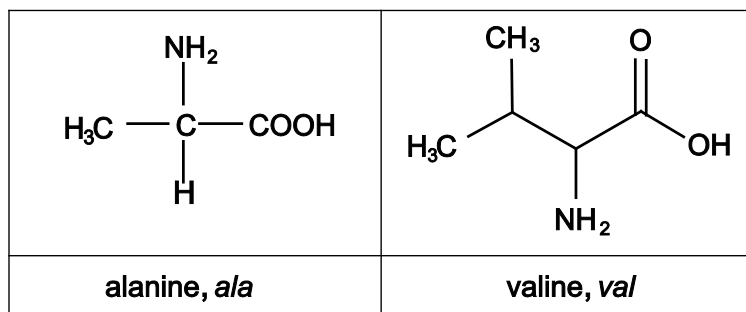
Question 4 (a) (i)–(iv)

4 Proteins are formed by the polymerisation of amino acids.

(a) (i) State the type of polymerisation used to form proteins.

..... [1]

(ii) The amino acids alanine and valine can be joined together to form a dipeptide.



Draw the structure of the dipeptide ala-val.

[1]

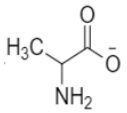
(iii) Draw the zwitterion of valine.

[1]

(iv) Draw the species formed by alanine at high pH.

[1]

Mark Scheme

4(a)(i)	Condensation	[1]	
4(a)(ii)	$\text{NH}_2\text{-CH}(\text{CH}_3)\text{-CO-NH-CH}(\text{C}_3\text{H}_7)\text{-COOH}$	[1]	Accept: val -ala not polymers
4(a)(iii)		[1]	allow -CO_2^-
4(a)(iv)		[1]	

Example candidate response—grade A

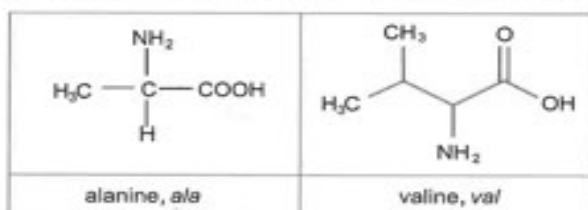
4 Proteins are formed by the polymerisation of amino acids.

(a) (i) State the type of polymerisation used to form proteins.

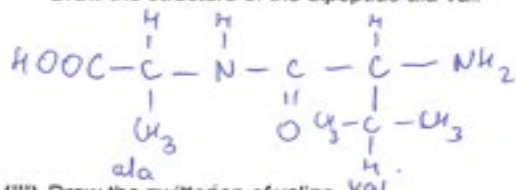
condensation

[1]

(ii) The amino acids alanine and valine can be joined together to form a dipeptide.

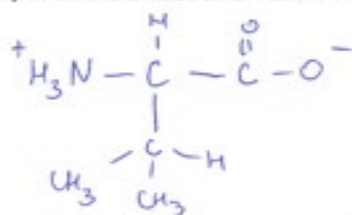


Draw the structure of the dipeptide ala-val.



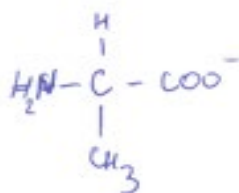
[1]

(iii) Draw the zwitterion of valine.



[1]

(iv) Draw the species formed by alanine at high pH.



[1]

For
Examiner's
Use

Examiner's comment

The candidate correctly names the type of polymerisation and deduces the structure of the dipeptide.

The representations of the zwitterion and the species under basic conditions are also given correctly.

Example candidate response—grade C

4 Proteins are formed by the polymerisation of amino acids.

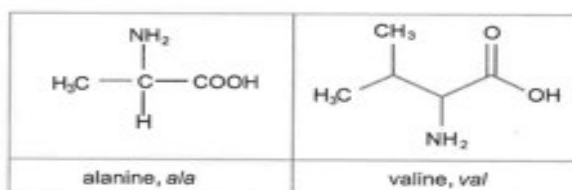
(a) (i) State the type of polymerisation used to form proteins.

addition polymerisation

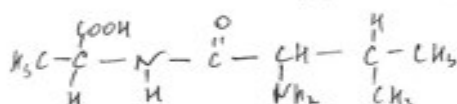
[1]

For
Examiner's
Use

(ii) The amino acids alanine and valine can be joined together to form a dipeptide.

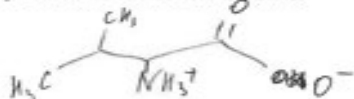


Draw the structure of the dipeptide ala-val.



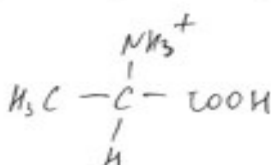
[1]

(iii) Draw the zwitterion of valine.



[1]

(iv) Draw the species formed by alanine at high pH.



[1]

Examiner's comment

The candidate incorrectly names the type of polymerisation but correctly writes the structure of the dipeptide.

The zwitterion is correctly represented, but the species under basic conditions is incorrect, as the candidate instead draws the species present in acidic conditions.

Example candidate response—grade E

4 Proteins are formed by the polymerisation of amino acids.

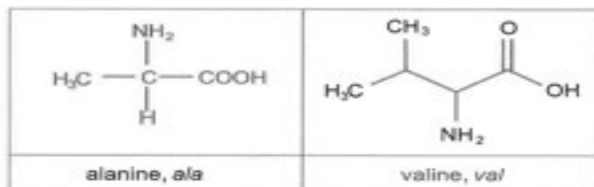
(a) (i) State the type of polymerisation used to form proteins.

addition polymerisation

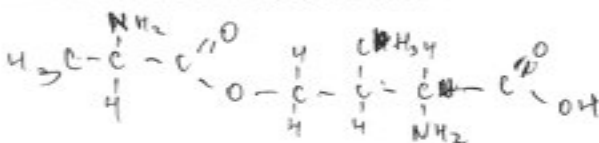
[1]

For
Examiner's
Use

(ii) The amino acids alanine and valine can be joined together to form a dipeptide.

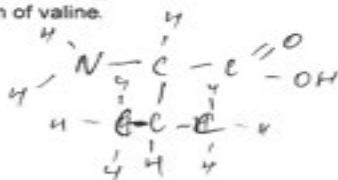


Draw the structure of the dipeptide ala-val.



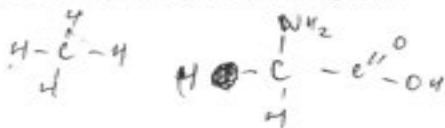
[1]

(iii) Draw the zwitterion of valine.



[1]

(iv) Draw the species formed by alanine at high pH.



[1]

Examiner's comment

The candidate incorrectly names the type of polymerisation and gives an incorrect structure for the dipeptide.

The structure of the zwitterion of valine is also incorrect, as no charges are shown.

The representation of the species under basic conditions is incorrect.

Question 4 (b)

(b) Proteins can form tertiary structure. Different forces and bonds stabilise the tertiary structure.

List **four** types of side-chain interactions and state the groups that are responsible for the formation of these interactions between amino acids residues.

.....

.....

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.....

.....

..... [3]

Mark Scheme

4(b)	✓ <u>ionic attractions/bonds/salt bridge</u> between CO_2^- and NH_3^+	1	Four correct answers -3 marks
	✓ <u>van der Waals/London dispersion/induced</u> between <u>alkyl / non-polar groups</u>	1	Three correct answers-2 marks
	✓ <u>hydrogen(H) bonding</u> between $\text{OH}/\text{NH}_2/\text{COOH}/\text{NH}$	1	Two correct answers -1 mark
	✓ <u>disulfide bridges</u> (<u>cysteine</u> residues/ S-S bonds)	[3]	List correct four types – 1 <u>mark</u>

Example candidate response—grade A

(b) Proteins can form tertiary structure. Different forces and bonds stabilise the tertiary structure.

List **four** types of side-chain interactions and state the groups that are responsible for the formation of these interactions between amino acids residues.

S-S : disulfide bridge ✓

N-H ; O-H ; polar hydrogen bonding ✓

R₁ --- R₂ : radial attraction / VWF ✓

Ionic bond : NH_3^+ --- COO^- ✓

.....

..... [3]

For
Examiner's
Use

3

Examiner's comment

The candidate identifies all four types of intermolecular interactions and gains 1 mark.

An additional 1 mark is awarded for each of the two correctly indicated interaction types with appropriate examples.

Example candidate response—grade C

(b) Proteins can form tertiary structure. Different forces and bonds stabilise the tertiary structure.

List four types of side-chain interactions and state the groups that are responsible for the formation of these interactions between amino acids residues.

Hydrogen bonds: between OH from carb. acid and H from NH_2 X

Van der Waals interaction: between R-groups, $\text{CH}_2 - \text{CH}_2$ ✓

Sulfide interaction: between sulphur (S) and CH_2 ✓

Ionic bonds: between positively charged and negatively charged ions ✓

[3] 2

For Examiner's Use

Examiner's comment

The candidate shows all four types of interactions but indicates specific groups for only two of them (van der Waals and sulfide interaction).

Example candidate response—grade E

(b) Proteins can form tertiary structure. Different forces and bonds stabilise the tertiary structure.

List four types of side-chain interactions and state the groups that are responsible for the formation of these interactions between amino acids residues.

Hydrogen bond - carboxylic and amino groups X

Sulfur bridge - S_2 X

Wander Wools Forces X

[3] 0

For Examiner's Use

Examiner's comment

The candidate does not list all four types of interactions; therefore, no marks are awarded. They also do not identify the corresponding functional groups responsible for the interactions.

Question 4 (c)

(c) Valine exists as a pair of stereoisomers.

(i) Explain the meaning of the term *stereoisomers*.

.....

.....

..... [2]

(ii) Explain how the pair of stereoisomers of valine can be distinguished.

.....

.....

..... [2]

Mark Scheme

4(c)(i)	Molecules with the <u>same structure/structural formula</u> ;	1	
	But with bonds/atoms/groups <u>arranged differently in space</u> ;	1	
		[2]	
4(c)(ii)	Plane <u>polarised</u> light;	1	OWTTE
	Will be rotated in opposite directions;	1	
		[2]	

Example candidate response—grade A

(c) Valine exists as a pair of stereoisomers.

(i) Explain the meaning of the term *stereoisomers*.

is the type of isomerism when the substance has a same structural formula but elements are differently arranged in space.

..... [2]

(ii) Explain how the pair of stereoisomers of valine can be distinguished.

The pair of isomers are mirrored for each other and when the polarized light beam is directed to isomers, they rotate in different clock directions.

..... [2]

Examiner's comment

The candidate correctly explains the meaning of stereoisomers using key phrases such as “same structural formula” and “differently arranged in space.”

A correct explanation is also given for how stereoisomers can be distinguished.

Example candidate response—grade C

(c) Valine exists as a pair of stereoisomers.

(i) Explain the meaning of the term stereoisomers.

Stereoisomers are isomers that
~~are cannot be~~ have different bond
placement in 3 dimensional systems.

[2]

(ii) Explain how the pair of stereoisomers of valine can be distinguished.

Ray of polarised light should
incident on valine and each
stereoisomer will rotate in diffe
rent directions.

[2]

Examiner's comment

The candidate does not explain the meaning of stereoisomers.

However, they correctly describe how to distinguish them, using terms such as “polarised light” and “rotate differently.”

Example candidate response—grade E

(c) Valine exists as a pair of stereoisomers.

(i) Explain the meaning of the term stereoisomers.

Stereoisomers - type of isomerism where the
isomers have same molecular formula but
different arrangement of atoms.

[2]

(ii) Explain how the pair of stereoisomers of valine can be distinguished.

Stereoisomers of valine can be distinguished by
different arrangement of chiral carbon, they
produce 2 various optical isomers.

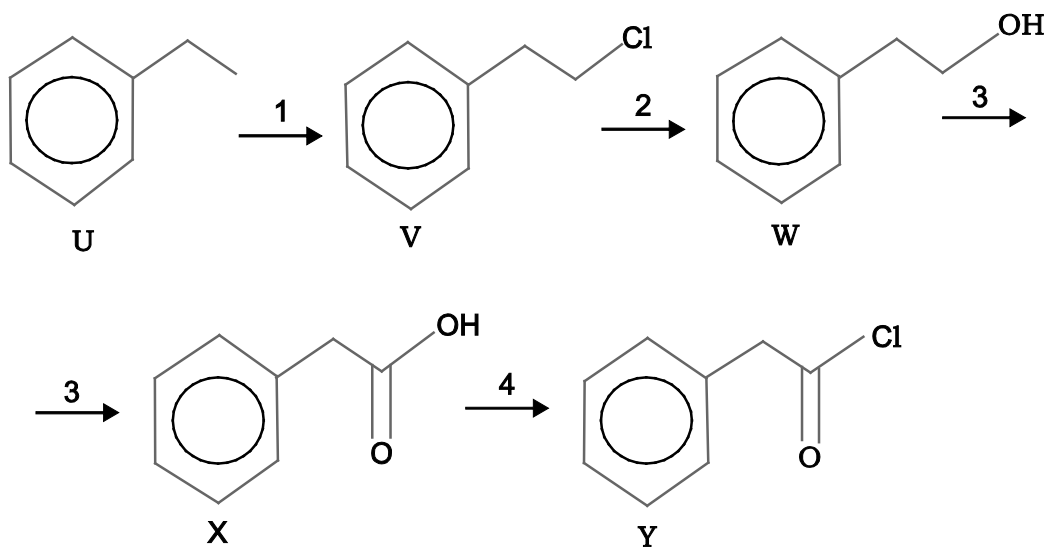
[2]

Examiner's comment

The candidate provides incorrect explanation for both the meaning of stereoisomers and the method used to distinguish them.

Question 5 (a), (b)

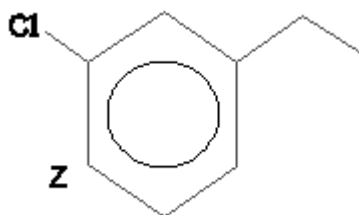
Phenylacetyl chloride, compound **Y**, is a colourless volatile liquid. It is used in manufacturing esters for flavourings. It can be synthesised from ethylbenzene, **U**, by the given route.



(a) Suggest reagents and conditions for Reaction 1.

..... [1]

(b) Give reagent and catalyst to produce compound **Z** (an isomer of **V**) from ethylbenzene.



..... [1]

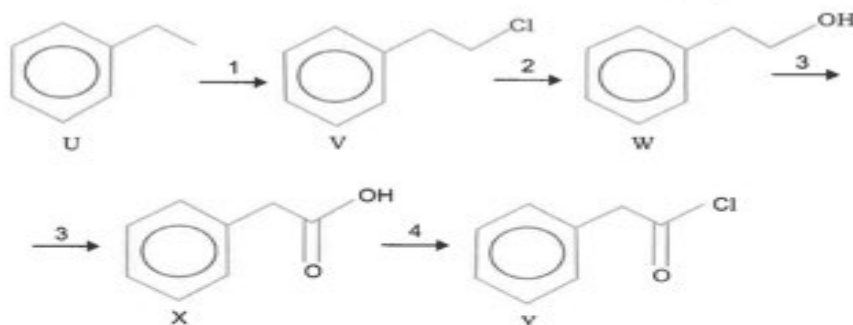
Mark Scheme

5(a)	Cl ₂ AND (UV) light/heat/ <u>hν</u>	[1]	Must be both answers to get mark
5(b)	Reagent Cl ₂ AND Catalyst AlCl ₃ /FeCl ₃ /Fe, etc.	[1]	Must be both answers to get mark

Example candidate response—grade A

- 5 Phenylacetyl chloride, compound Y, is a colourless volatile liquid. It is used in manufacturing esters for flavourings. It can be synthesised from ethylbenzene, U, by the given route.

For
Examiner's
Use



- (a) Suggest reagents and conditions for Reaction 1.

Reagents: Cl_2 + UV light [1] ✓

- (b) Give reagent and catalyst to produce compound Z (an isomer of V) from ethylbenzene.



Cl_2 and AlCl_3 [1] ✓

Examiner's comment

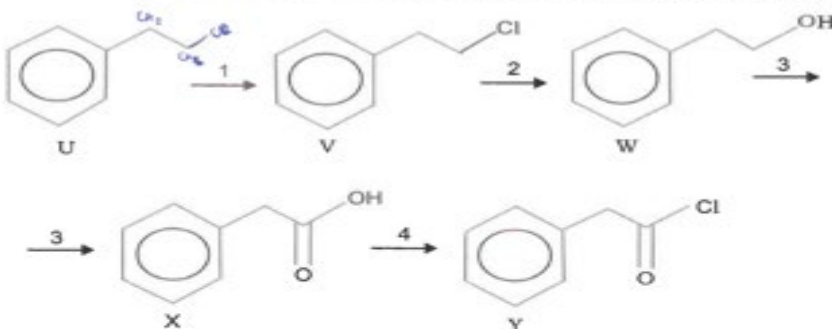
The candidate correctly states the reagent and conditions for Reaction 1 (Cl_2 and UV light), and also gives the correct reagent and catalyst (Cl_2 and AlCl_3) for the formation of compound Z.

Full marks are awarded for both parts.

Example candidate response—grade C

- 5 Phenylacetyl chloride, compound Y, is a colourless volatile liquid. It is used in manufacturing esters for flavourings. It can be synthesised from ethylbenzene, U, by the given route.

For
Examiner's
Use



- (a) Suggest reagents and conditions for Reaction 1.

Cl_2 , UV light free radical addition [1] ✓

- (b) Give reagent and catalyst to produce compound Z (an isomer of V) from ethylbenzene.



HCl , H_2SO_4 [1] ✗

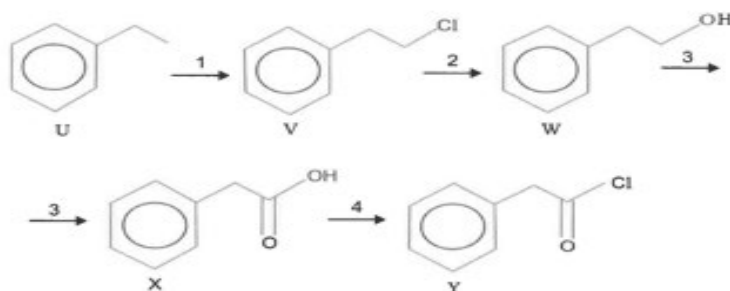
Examiner's comment

In part (a), the candidate correctly states the reagent and condition (Cl_2 and UV light) for the free-radical substitution reaction, therefore 1 mark is awarded.

In part (b), the reagent and catalyst provided (HCl and H_2SO_4) are incorrect for the formation of compound Z. No mark is awarded for this part.

Example candidate response—grade E

- 5 Phenylacetyl chloride, compound Y, is a colourless volatile liquid. It is used in manufacturing esters for flavourings. It can be synthesised from ethylbenzene, U, by the given route.



- (a) Suggest reagents and conditions for Reaction 1.

AlCl_3 , reflux, $\text{CH}_3\text{-Cl}$

[1]

- (b) Give reagent and catalyst to produce compound Z (an isomer of V) from ethylbenzene.



H_2SO_4 , reflux, $\text{CH}_3\text{-Cl}$

[1]

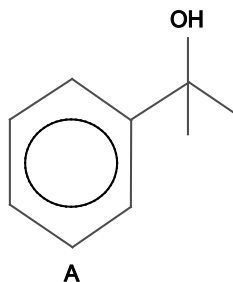
Examiner's comment

In part (a), the candidate incorrectly states the reagent and conditions. AlCl_3 and a reflux setup are not appropriate for Reaction 1, which requires Cl_2 and UV light for free-radical substitution. No mark is awarded for this part.

In part (b), the candidate gives H_2SO_4 and $\text{CH}_3\text{-Cl}$, which are not suitable for the formation of compound Z, therefore no mark is awarded.

Question 5 (c)

- (c) Suggest a test that can help to distinguish compound W from compound A.



Reagents

Observation with **W**

Observation with **A**

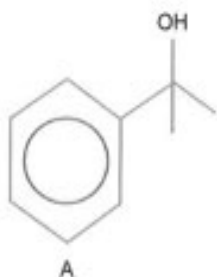
[3]

Mark Scheme

5(c)	M1 acidified potassium dichromate or $K_2Cr_2O_7 / H_2SO_4$ OR $K_2Cr_2O_7 / H^+$ OR <u>acidified</u> $K_2Cr_2O_7$ M2 (orange to) green solution OR goes green M3 (solution) remains orange or no reaction or no (observed) change Alternative using $KMnO_4 / H_2SO_4$ M1 acidified potassium <u>manganate</u>(VII) / potassium permanganate or $KMnO_4 / H_2SO_4$ OR $KMnO_4 / H^+$ OR acidified $KMnO_4$ M2 colourless solution OR goes colourless M3 (solution) remains purple or no reaction or no (observed) change	1 1 1 [3] If no reagent or incorrect reagent in M1, no marks for M1, M2 and M3 If incomplete / inaccurate attempt at reagent e.g. "dichromate" or "<u>dichromate</u>(IV)" or incorrect formula or no acid, penalise M1 only and mark on For M2 ignore dichromate described as "yellow" or "red" For M3 ignore "nothing (happens)" or "no observation" For M1 If incomplete / inaccurate attempt at reagent e.g. "manganate" or "<u>manganate</u>(IV)" or incorrect formula or no acid, penalise M1 only and mark on (Credit alkaline $KMnO_4$ for possible full marks but M2 gives <u>brown</u> precipitate
------	---	---

Example candidate response—grade A

(c) Suggest a test that can help to distinguish compound W from compound A.



Reagents

$K_2Cr_2O_7$ with H_2SO_4 ✓

Observation with W

colour change from orange to green ✓

Observation with A

no reaction takes place ✓

[3]

3

For
Examiner's
Use

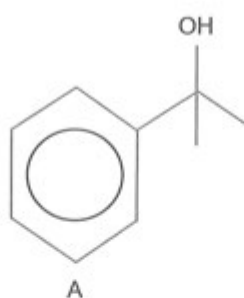
Examiner's comment

The candidate correctly states the reagent as acidified potassium dichromate and provides accurate observations for both compounds.

A clear colour change from orange to green is noted for compound W, indicating oxidation, while no reaction is observed with compound A.

Example candidate response—grade C

(c) Suggest a test that can help to distinguish compound W from compound A.



Reagents

$K_2Cr_2O_7$ ✓ x

Observation with W

changes colour from orange to green ✓

Observation with A

no change in colour ✓

[3]

For
Examiner's
Use

2

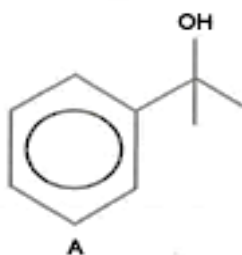
Examiner's comment

The candidate states an incorrect reagent, omitting the required acid (H_2SO_4) that should be used with potassium dichromate.

However, both observations are correct: a colour change from orange to green for compound W, and no change for compound A. Therefore, 2 out of 3 marks are awarded.

Example candidate response—grade E

(c) Suggest a test that can help to distinguish compound W from compound A.



Reagents

Tollen's reagent

X

Observation with W

silver mirror

X

Observation with A

no reaction

X

[3]

For
Examiner's
Use

0

Examiner's comment

The candidate incorrectly identifies Tollens' reagent for this test. This reagent is used to distinguish aldehydes.

As a result, the observations given for compound W (silver mirror) and compound A (no reaction) are also incorrect in this context.

Question 5 (d) (e)

(d) Give the name of the mechanism for Reaction 2.

..... [1]

(e) In order to ensure that the oxidation to phenylacetic acid, compound X, is complete, the reaction is carried out under reflux.

Describe what happens when a reaction mixture is refluxed and why it is necessary, in this case, for a complete oxidation to phenylacetic acid.

.....
.....
.....
..... [2]

Mark Scheme

5(d)	Nucleophilic substitution	[1]	both words required. Accept phonetic spelling
5(e)	A mixture of liquids is heated AND <u>Vapour is formed condensed back/return</u> To ensure that any alcohol/aldehyde is completely oxidised .	1 1 [2]	OWTTE

Example candidate response—grade A

(d) Give the name of the mechanism for Reaction 2.

nucleophilic substitution ✓ ✓ [1]

(e) In order to ensure that the oxidation to phenylacetic acid, compound X, is complete, the reaction is carried out under reflux.

Describe what happens when a reaction mixture is refluxed and why it is necessary, in this case, for a complete oxidation to phenylacetic acid.

it will be fully oxidised when
refluxed because formal gaseous compounds are
condensed and can be used and heat again
This is very important because without reflux it may
oxidise only to aldehyde. ✓ ✓ [2]

Examiner's comment

The candidate correctly identifies the name of the mechanism as nucleophilic substitution.

They also correctly deduce the intermediate oxidation product and demonstrate an understanding of the purpose of reflux, referring to the idea of “heating again” to ensure complete oxidation.

Example candidate response—grade C

- (d) Give the name of the mechanism for Reaction 2.

nucleophilic substitution ✓ [1] 1

- (e) In order to ensure that the oxidation to phenylacetic acid, compound X, is complete, the reaction is carried out under reflux.

Describe what happens when a reaction mixture is refluxed and why it is necessary, in this case, for a complete oxidation to phenylacetic acid.

the reaction of oxidation has intermediate products as
aldehydes and to make sure that all alcohol turns into acid ✓
~~acid then used in manufacturing of acyl chlorides~~
and reflux. Formed acid then used in manufacturing [2] 1
acyl chlorides ✗

Examiner's comment

The candidate correctly identifies the name of the mechanism as nucleophilic substitution and correctly deduces the intermediate oxidation product as an aldehyde.

However, the response does not clearly explain the purpose of reflux or what happens during the process, resulting in an incomplete explanation.

Example candidate response—grade E

- (d) Give the name of the mechanism for Reaction 2.

electrophilic substitution ✗ [1] 0

- (e) In order to ensure that the oxidation to phenylacetic acid, compound X, is complete, the reaction is carried out under reflux.

Describe what happens when a reaction mixture is refluxed and why it is necessary, in this case, for a complete oxidation to phenylacetic acid.

Reflux is necessary to produce electrophile, which
will react with phenylacetic acid, for the equilibrium
to occur and make sure that all reactants react
fully, increase the rate of reaction ✗ [2] 0

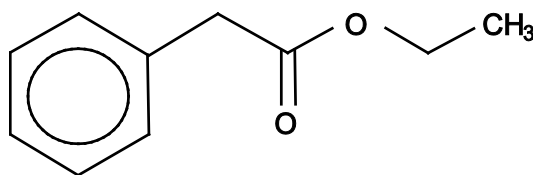
Examiner's comment

In part (d), the candidate identifies the general type of reaction as substitution but gives the incorrect mechanism name (electrophilic substitution) for this context. Since the full and correct name — nucleophilic substitution — is required, no mark is awarded.

In part (e), the candidate does not provide a chemically accurate explanation of what happens during reflux or why it is necessary for complete oxidation.

Question 5 (f) (i), (ii)

(f) Phenylacetyl chloride, compound **Y**, reacts with ethanol to produce ester **B**.



B

- (i) Name the mechanism for the reaction between phenylacetyl chloride and ethanol.

..... [1]

- (ii) Outline a mechanism for this reaction. You can represent aryl group as "R". Include all necessary curly arrows, lone pairs and relevant dipoles.

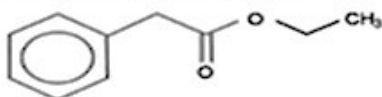
[4]

Mark Scheme

5(f)(i)	(nucleophilic) addition-elimination	[1]	Not electrophilic addition-elimination Ignore esterification
5(f)(ii)	structure (1) M3 3 arrows (1) M4 M1 arrow from the oxygen O-H of the alcohol group to the carbon atom of the carbonyl group and arrow from the <u>double bond</u> C=O group to the oxygen of the carbonyl group M2 Formation of a tetrahedral intermediate in which carbon is bonded to the oxygen of ethanol, chlorine and oxygen of the carbonyl group M3 final formula of the ester M4 3 arrows in the intermediate	1 1 1 1 [4]	$\delta+$ on C=O loses M1 If Cl lost with C=O breaking, max1 for M1 do not accept M3 without M1/M2

Example candidate response—grade A

(f) Phenylacetyl chloride, compound Y, reacts with ethanol to produce ester B.

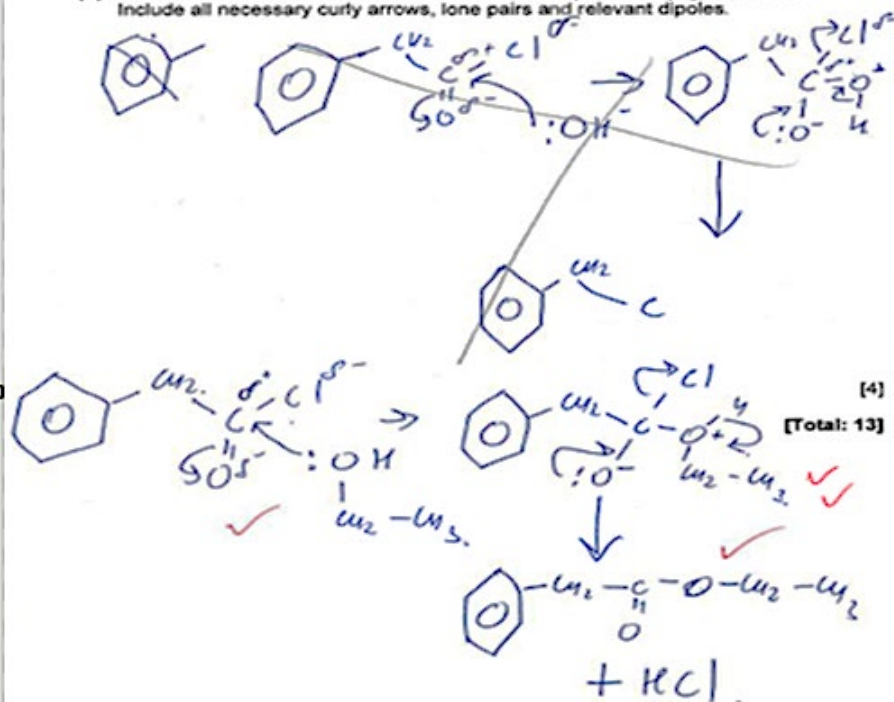


B

(i) Name the mechanism for the reaction between phenylacetyl chloride and ethanol.

nucleophilic addition-elimination [1]

(ii) Outline a mechanism for this reaction. You can represent aryl group as "R". Include all necessary curly arrows, lone pairs and relevant dipoles.



Examiner's comment

The candidate correctly identifies the name of the mechanism.

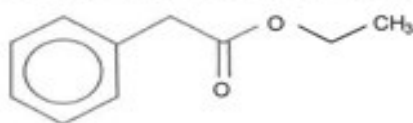
The curly arrow from the lone pair on the oxygen to the partially positive carbon atom is accurately drawn, as is the arrow from the carbonyl double bond to the oxygen.

The tetrahedral intermediate is correctly represented with all three arrows, and the final product is accurately drawn.

Full marks are awarded.

Example candidate response—grade C

(f) Phenylacetyl chloride, compound Y, reacts with ethanol to produce ester B.

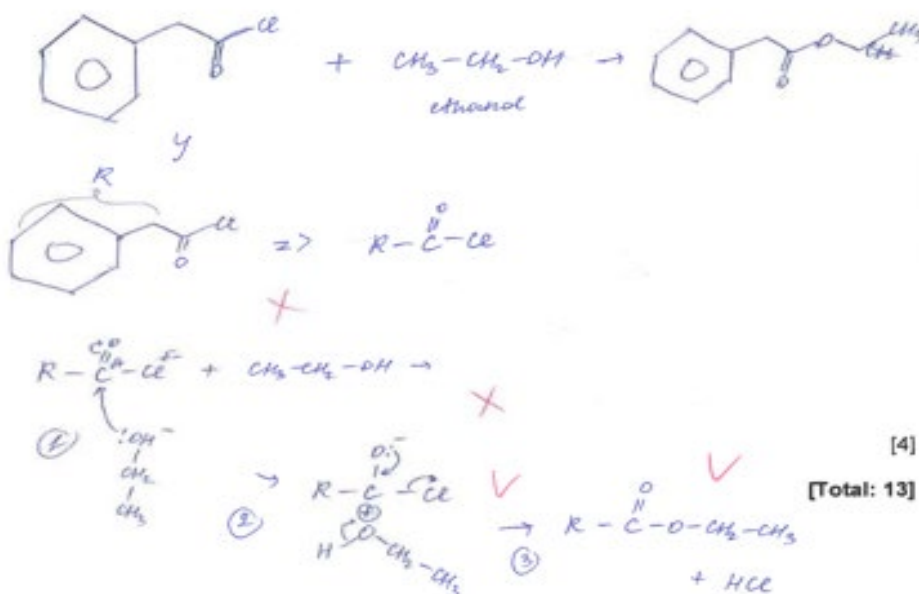


B

(i) Name the mechanism for the reaction between phenylacetyl chloride and ethanol.

(esterification) nucleophilic addition-elimination mechanism ✓ [1]

(ii) Outline a mechanism for this reaction. You can represent aryl group as "R". Include all necessary curly arrows, lone pairs and relevant dipoles.



Examiner's comment

The candidate correctly identifies the name of the mechanism.

The arrow from the lone pair on the oxygen to the partially positive carbon atom and the arrow from the carbonyl double bond to the oxygen are both correctly shown. However, a charge is incorrectly placed on the alcohol group, therefore, no mark is awarded for this part.

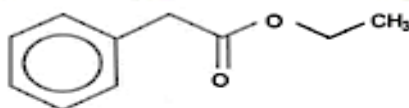
The tetrahedral intermediate is accurately represented with all three curly arrows.

The final product is incorrectly drawn, with the $\text{-CH}_2\text{-}$ group missing.

Two out of 4 marks are awarded.

Example candidate response—grade E

- (f) Phenylacetyl chloride, compound Y, reacts with ethanol to produce ester B.

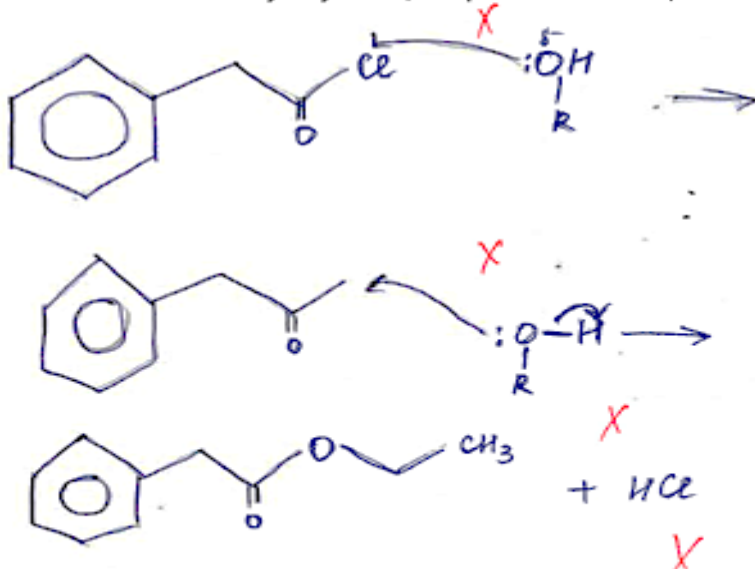


B

- (i) Name the mechanism for the reaction between phenylacetyl chloride and ethanol.

esterification, substitution [1] X

- (ii) Outline a mechanism for this reaction. You can represent any group as "R". Include all necessary curly arrows, lone pairs and relevant dipoles.



[4]

[Total: 13]

For
Examiner's
Use

0

0

3

Examiner's comment

The candidate gives an incorrect name for the mechanism.

The arrow from the lone pair on the oxygen to the chlorine atom is shown incorrectly, and the arrow to the final product is also incorrect.

The tetrahedral intermediate is missing, along with the required three curly arrows.

Although the final product is drawn, no marks are awarded for it, as marks for M1, M2, and M4 are not earned.

Question 6 (a)

6 In this question, give all your answers in **three** significant figures.

- (a) Zinc nitrate decomposes on heating to form zinc oxide, nitrogen (IV) oxide and oxygen.

- (i) Write a balanced chemical equation for this decomposition reaction. Include state symbols.

..... [2]

- (ii) Thermal decomposition of a sample of zinc nitrate produced 2.16 g of zinc oxide.

Calculate the amount, in moles, of zinc oxide in 2.16 g.

amount = mol [1]

- (iii) Calculate the total amount, in moles, of gas produced from this sample of zinc nitrate.

amount = mol [1]

Mark Scheme

6(a)(i)	$2\text{Zn}(\text{NO}_3)_2(\text{s}) \rightarrow 2\text{ZnO}(\text{s}) + 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$ correct formulae AND state symbols correct balancing	1 1 [2]	
6(a)(ii)	$2.16/81.4 = 0.0265$ Penalise if not 3 sig figs in this clip only	[1]	Do not accept 81
6(a)(iii)	$0.0265 \times 2.5 = 0.0663$	[1]	Ecf

Example candidate response—grade A

6 In this question, give all your answers in **three** significant figures.

(a) Zinc nitrate decomposes on heating to form zinc oxide, nitrogen (IV) oxide and oxygen.

(i) Write a balanced chemical equation for this decomposition reaction. Include state symbols.

$2\text{Zn}(\text{NO}_3)_2(\text{s}) \rightarrow 2\text{ZnO}(\text{s}) + 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$ [2] ✓

(ii) Thermal decomposition of a sample of zinc nitrate produced 2.16 g of zinc oxide.
Calculate the amount, in moles, of zinc oxide in 2.16 g.

$n = \frac{m}{M_r} = \frac{2.16}{81.4} = 0.0265$

amount = 0.0265 mol [1] ✓

(iii) Calculate the total amount, in moles, of gas produced from this sample of zinc nitrate.

$0.0265 \cdot 2 + 0.0265 \cdot \frac{1}{2} = 0.0663$

amount = 0.0663 mol [1] ✓

For Examiner's Use

2

1

1

Examiner's comment

The candidate correctly writes the formulae of the substances and the physical state of each. The chemical equation is also correctly balanced. The candidate is awarded full marks (2 marks).

In part (a)(ii), they correctly use the atomic mass of zinc from the data booklet, and the final answer is given to three significant figures. The candidate is awarded 1 mark.

In part (a)(iii), using the chemical equation they correctly calculate the total number of moles for both gases. The candidate is awarded 1 mark.

Example candidate response—grade C

6 In this question, give all your answers in **three** significant figures.

(a) Zinc nitrate decomposes on heating to form zinc oxide, nitrogen (IV) oxide and oxygen.

(i) Write a balanced chemical equation for this decomposition reaction. Include state symbols.



(ii) Thermal decomposition of a sample of zinc nitrate produced 2.16 g of zinc oxide.

Calculate the amount, in moles, of zinc oxide in 2.16 g.

$$n = \frac{m}{M_r} = \frac{2.16}{81.4 \text{ g mol}^{-1}} \approx 0.03 \text{ mol} \quad X$$

$$M_r(\text{ZnO}) = 65.4 + 16 = 81.4 \text{ g mol}^{-1}$$

amount = 0.03 mol [1]

(iii) Calculate the total amount, in moles, of gas produced from this sample of zinc nitrate.

$$1 - 2 \quad X = \frac{0.03 \cdot 2}{1} = 0.06 \text{ mol}$$

amount = 0.06 mol [1]

For
Examiner's
Use

2

0

0

Examiner's comment

The candidate correctly writes the formulae of the substances and the physical state of each. The chemical equation is also correctly balanced. The candidate receives full marks (2 marks). In part (a)(ii), they correctly use the atomic mass of zinc, but the number of moles is not given to three significant figures. The value is approximated. Therefore, no mark is awarded for this part.

In part (a)(iii), they calculate the number of moles for only one gas instead of both. Therefore, no mark is awarded for this part.

Example candidate response—grade E

6 In this question, give all your answers in **three** significant figures.

(a) Zinc nitrate decomposes on heating to form zinc oxide, nitrogen (IV) oxide and oxygen.

(i) Write a balanced chemical equation for this decomposition reaction. Include state symbols.



(ii) Thermal decomposition of a sample of zinc nitrate produced 2.16 g of zinc oxide.

Calculate the amount, in moles, of zinc oxide in 2.16 g.

$$n = \frac{2.16}{806} = 2.01 \cdot 10^{-3}$$

amount = $6.02 \cdot 10^{-3}$ mol [1]

(iii) Calculate the total amount, in moles, of gas produced from this sample of zinc nitrate.

$$0.0214 + 0.0214 + \frac{6.02 \cdot 10^{-3}}{2} = 0.045175 \text{ mol}$$

amount = 0.045175 mol [1]

For
Examiner's
Use

0

0

0

Examiner's comment

The candidate writes incorrect formulae and does not correctly balance the chemical equation. No marks are awarded for this part.

In part (a)(ii), the calculation of the number of moles of zinc oxide is incorrect. No mark is awarded.

In part (a)(iii), the number of moles for both gases is calculated incorrectly. No mark is awarded for this part either.

Question 6 (b)

- (b) In another experiment, a different sample of zinc nitrate decomposed to produce 0.506 mol of gas. Calculate the volume, in dm^3 , that this gas would occupy at 357 K and $1.00 \times 10^5 \text{ Pa}$ (The gas constant $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$).

volume = dm^3 [2]

Mark Scheme

6(b)	M1: $pV = nRT$	1	However if 15.0 m^3 or cm^3 allow 1 (ie penalise wrong units in final answer)
	$V = \frac{0.506 \times 8.31 \times 357}{100\,000} = 0.0150 \text{ m}^3$	1	
	15.0 (dm^3) All 2 marks for 15.0 (dm^3)	[2]	

Example candidate response—grade A

- (b) In another experiment, a different sample of zinc nitrate decomposed to produce 0.506 mol of gas. Calculate the volume, in dm^3 , that this gas would occupy at 357 K and $1.00 \times 10^5 \text{ Pa}$ (The gas constant $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$).

$$pV = nRT \quad \checkmark$$

$$V = \frac{nRT}{p}$$

$$V = \frac{0.506 \times 8.31 \times 357}{10^5} = 0.015 \text{ m}^3$$

$$V = 15.0 \text{ dm}^3$$

volume = 15.0 dm^3 [2] 2

Examiner's comment

The candidate correctly uses the ideal gas equation and provides the volume with the correct unit. The candidate is awarded full marks.

Example candidate response—grade C

(b) In another experiment, a different sample of zinc nitrate decomposed to produce 0.506 mol of gas. Calculate the volume, in dm^3 , that this gas would occupy at 357 K and $1.00 \times 10^5 \text{ Pa}$ (The gas constant $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$).

$$pV = nRT \quad \checkmark$$
$$V = \frac{nRT}{p} = \frac{0.506 \cdot 357 \cdot 8.31}{10^5} = 0.0150$$

volume = 0.0150 dm^3 [2]

Examiner's comment

The candidate correctly applies the ideal gas equation and substitutes the values accurately. However, the final answer is given in the wrong unit. Therefore, the candidate is awarded 1 mark.

Example candidate response—grade E

(b) In another experiment, a different sample of zinc nitrate decomposed to produce 0.506 mol of gas. Calculate the volume, in dm^3 , that this gas would occupy at 357 K and $1.00 \times 10^5 \text{ Pa}$ (The gas constant $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$).

$$\frac{0.506 \cdot 8.31}{8.31 \cdot 357} = 12.81 \quad \times$$

volume = dm^3 [2]

Examiner's comment

The candidate incorrectly applies the ideal gas equation and miscalculates the volume. No marks are awarded for this part.

Question 6 (c)

(c) 0.0247 mol sample of zinc oxide, produced from the decomposition of zinc nitrate, was reacted with nitric acid.

(i) Write a balanced chemical equation for this reaction.

..... [1]

(ii) Calculate the amount, in moles, of acid needed to react completely with the 0.0247 mol sample of zinc oxide.

amount = mol [1]

(iii) 0.0247 mol sample of zinc oxide needed 45.7 cm³ of nitric acid for complete reaction. Use this information and your answer to part (c)(ii) to calculate the concentration, in mol dm⁻³, of the nitric acid.

amount = mol dm⁻³ [1]

Mark Scheme

6(c)(i)	$\text{ZnO} + 2\text{HNO}_3 = \text{Zn}(\text{NO}_3)_2 + \text{H}_2\text{O}$	[1]	Must be balanced
6(c)(ii)	$0.0247 \times 2 = 0.0494$	[1]	ecf Allow 0.0500
6(c)(iii)	1.08 mol dm^{-3}	[1]	Ecf

Example candidate response—grade A

- (c) 0.0247 mol sample of zinc oxide, produced from the decomposition of zinc nitrate, was reacted with nitric acid.



- (i) Write a balanced chemical equation for this reaction.



- (ii) Calculate the amount, in moles, of acid needed to react completely with the 0.0247 mol sample of zinc oxide.



0.0247 mol ?

$$\begin{array}{l} 1 - 0.0247 \text{ mol} \\ 2 - x \end{array} \quad \left| \quad x = 0.0494 \text{ mol (HNO}_3\text{)} \right.$$

amount = 0.0494 mol [1]

- (iii) 0.0247 mol sample of zinc oxide needed 45.7 cm³ of nitric acid for complete reaction. Use this information and your answer to part (c)(ii) to calculate the concentration, in mol dm⁻³, of the nitric acid.

$$C = \frac{n}{V}$$

$$C = \frac{0.0494}{\frac{45.7}{1000}} = 1.08 \text{ mol/dm}^3$$

amount = 1.08 mol dm⁻³ [1]

Examiner's comment

In part (c)(i), the candidate correctly writes the balanced chemical equation.

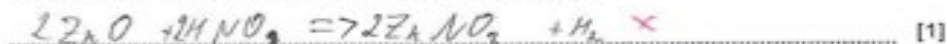
In part (c)(ii), the candidate accurately calculates the amount (in moles) of nitric acid needed using the correct mole ratio from the equation.

In part (c)(iii), the candidate correctly applies the concentration formula and uses the correct values to calculate the concentration in mol dm⁻³.

Example candidate response—grade C

- (c) 0.0247 mol sample of zinc oxide, produced from the decomposition of zinc nitrate, was reacted with nitric acid.

(i) Write a balanced chemical equation for this reaction.



(ii) Calculate the amount, in moles, of acid needed to react completely with the 0.0247 mol sample of zinc oxide.

$$2 : 2$$

$$n(ZnO) = 0,0247$$

amount = $0,0247$ ^{ecf} mol [1]

(iii) 0.0247 mol sample of zinc oxide needed 45.7 cm³ of nitric acid for complete reaction. Use this information and your answer to part (c)(ii) to calculate the concentration, in mol dm⁻³, of the nitric acid.

$$[HNO_3] = \frac{0,0247}{0,0457} = 0,540 \frac{\text{mol}}{\text{dm}^3}$$

amount = $0,540$ ^{ecf} mol dm⁻³ [1]

For
Examiner's
Use

0

1

1

Examiner's comment

The candidate incorrectly balances the chemical equation in part (c)(i); therefore, no mark is awarded.

However, the candidate is awarded 2 marks in parts (c)(ii) and (c)(iii) through error carried forward (ecf), demonstrating correct method and calculations based on their previous answer.

Example candidate response—grade E

- (c) 0.0247 mol sample of zinc oxide, produced from the decomposition of zinc nitrate, was reacted with nitric acid.

For
Examiner's
Use

- (i) Write a balanced chemical equation for this reaction.



- (ii) Calculate the amount, in moles, of acid needed to react completely with the 0.0247 mol sample of zinc oxide.

~~$0,247 \times 81,4 = 17,0$~~
 $0,247 \cdot 81,4 = 17 \cdot x$
 $x = 1,18$

1,18
 amount = ~~0,0546~~ mol [1] X 0.

- (iii) 0.0247 mol sample of zinc oxide needed 45.7 cm³ of nitric acid for complete reaction. Use this information and your answer to part (c)(ii) to calculate the concentration, in mol dm⁻³, of the nitric acid.

$n = cV$ $c = \frac{n}{V}$ $V = x$ $V = 9,57 \cdot 10^{-4}$
 $0,0247$
 $x = \frac{1,18}{9,57 \cdot 10^{-4}}$
 $x = \frac{1,18 \cdot 45,7}{0,01}$

25,8
 amount = ~~0,0546~~ mol dm⁻³ [1] X 0.

Examiner's comment

The candidate incorrectly balances the chemical equation in part (c)(i); therefore, no marks are awarded.

In parts (c)(ii) and (c)(iii), the candidate incorrectly calculates both the number of moles and the concentration. No marks are awarded for these parts either.

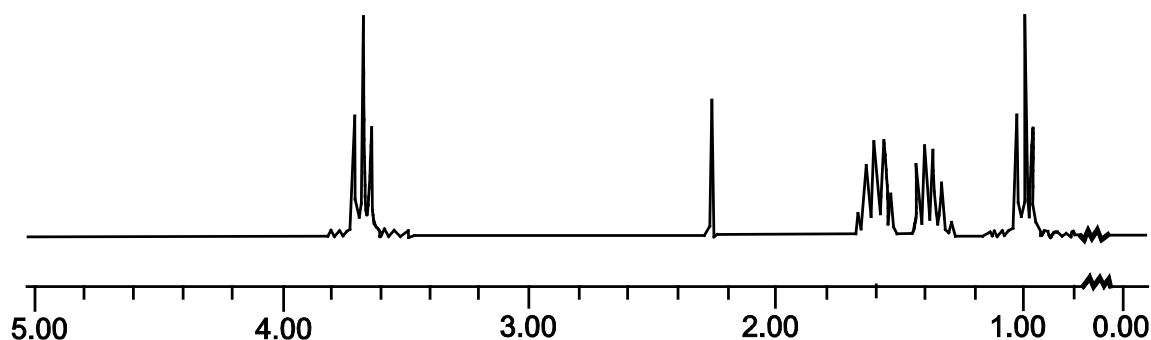
Question 7 (a), (b)

Isomers of butanol have comparatively better fuel properties and are less corrosive compared to ethanol. One of the isomers of butanol, **X**, is analysed by IR and $^1\text{H-NMR}$ spectroscopy and data is collected for further experiments.

(a) Suggest how IR spectrum helps the student to determine presence of alcohol group.

..... [1]

(b) The $^1\text{H-NMR}$ spectrum of isomer **X** is shown below.



(i) Deduce the number of different types of protons present in a compound.

..... [1]

(ii) The spectrum shows two triplets with different chemical shifts. Identify groups that are responsible for each chemical shift.

..... [1]

(iii) Explain the reason why peak at $\delta = 3.63$ shifts more to the left.

..... [1]

(iv) Deduce the structural formula of an isomer **X**.

[1]

Mark Scheme

7(a)	Strong absorption at 3600-3640\3200-3640cm ⁻¹	[1]	
7 (b) (i)	5	[1]	
7 (b) (ii)	Peak at $\delta = 3.63$ is -CH ₂ next to OH/ -CH-O- / -CH ₂ -O- AND Peak at $\delta = 0.97$ is CH ₃ next to CH ₂ / -CH-CH ₂ - / -CH ₃ -CH ₂	[1]	
7 (b) (iii)	-CH ₂ group is connected <u>closer</u> to more <u>electronegative</u> element/oxygen/more polar	[1]	
7 (b) (iv)	CH ₃ -CH ₂ -CH ₂ -CH ₂ -OH	[1]	

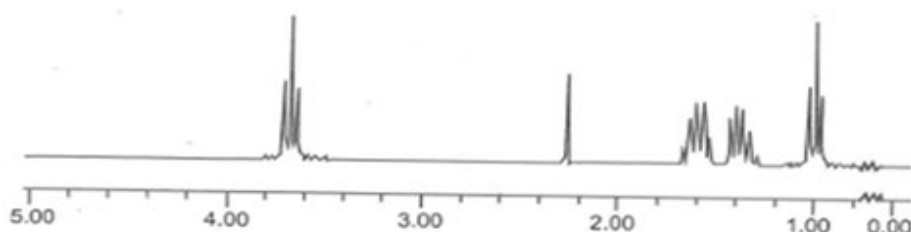
Example candidate response—grade A

7 Isomers of butanol have comparatively better fuel properties and are less corrosive compared to ethanol. One of the isomers of butanol, X, is analysed by IR and ¹H-NMR spectroscopy and data is collected for further experiments.

(a) Suggest how IR spectrum helps the student to determine presence of alcohol group.

it will have a strong peak at 3600-3670 cm⁻¹ wavenumber and a strong peak at 1050-1300 cm⁻¹ wavenumber [1]

(b) The ¹H-NMR spectrum of isomer X is shown below.



(i) Deduce the number of different types of protons present in a compound.

5 [1]

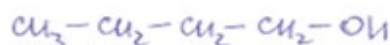
(ii) The spectrum shows two triplets with different chemical shifts. Identify groups that are responsible for each chemical shift.

the triplet at ~1 ppm is CH₃ next to CH₂
the triplet at ~3.63 ppm is CH₂ between CH₂ and OH [1]

(iii) Explain the reason why peak at $\delta = 3.63$ shifts more to the left.

because it is closer to an electronegative atom O in -OH group [1]

(iv) Deduce the structural formula of an isomer X.



[1]

Examiner's comment

The candidate correctly identifies the IR absorption range for the O-H bond (3600–3640 cm⁻¹), accurately states that there are five proton environments, correctly assigns the two triplets to -CH₂- groups adjacent to other -CH₂- groups, applying the (n+1) rule, and deduces and proposes the correct structure of butan-1-ol.

Precise IR range, clear NMR interpretation, and a logical structure demonstrate excellent application of the data booklet and splitting rules.

Example candidate response—grade C

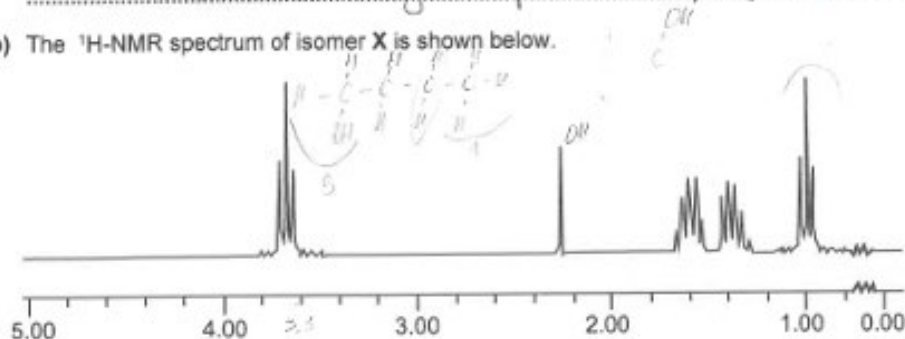
- 7 Isomers of butanol have comparatively better fuel properties and are less corrosive compared to ethanol. One of the isomers of butanol, X, is analysed by IR and ^1H -NMR spectroscopy and data is collected for further experiments.

For
Examiner's
Use

- (a) Suggest how IR spectrum helps the student to determine presence of alcohol group.

it would give specific frequency X [1] 0

- (b) The ^1H -NMR spectrum of isomer X is shown below.



- (i) Deduce the number of different types of protons present in a compound.

5 ✓ [1] 1

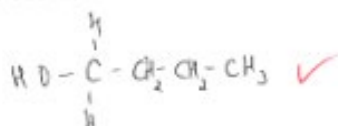
- (ii) The spectrum shows two triplets with different chemical shifts. Identify groups that are responsible for each chemical shift.

$-\text{O}-\text{CH}_2$ and CH_2-C X [1] 0

- (iii) Explain the reason why peak at $\delta = 3.63$ shifts more to the left.

because the proton at this environment is closer to highly electronegative atom ✓ [1] 1

- (iv) Deduce the structural formula of an isomer X.



[1] 1

Examiner's comment

The candidate does not state the IR absorption range from the Data Booklet; therefore, no mark is awarded for this part. The candidate correctly identifies five proton environments. There is no detailed explanation for each triplet, therefore, no mark is awarded, but the explanation for the peak at 3.63 ppm is correct. The structure of butan-1-ol is correctly deduced.

Example candidate response—grade E

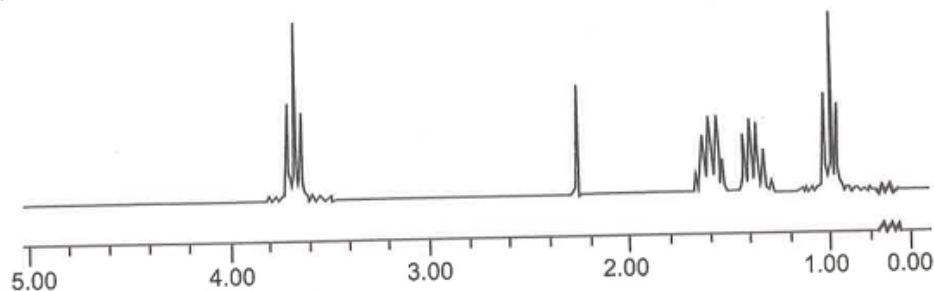
- 7 Isomers of butanol have comparatively better fuel properties and are less corrosive compared to ethanol. One of the isomers of butanol, X, is analysed by IR and ^1H -NMR spectroscopy and data is collected for further experiments.

For
Examiner's
Use

- (a) Suggest how IR spectrum helps the student to determine presence of alcohol group.

because of wavenumber of IR passed through the solution [1] D

- (b) The ^1H -NMR spectrum of isomer X is shown below.



- (i) Deduce the number of different types of protons present in a compound.

$\text{CH}_3 - \text{C} - \text{CH}_2 - \text{C} - \text{OH}$ [1] X

- (ii) The spectrum shows two triplets with different chemical shifts. Identify groups that are responsible for each chemical shift.

$\text{C} - \text{CH}_2 - \text{C} \sim 2.4$
 $-\text{C} - \text{OH} \sim 3.6$ [1] X

- (iii) Explain the reason why peak at $\delta = 3.63$ shifts more to the left.

because it is a functional group and it
shifts more [1] X

- (iv) Deduce the structural formula of an isomer X.

$\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{OH}$ [1] X

Examiner's comment

The candidate does not use information from the Data Booklet and the H-NMR spectrum effectively. The response is vague and lacks scientific precision. A correct explanation should reference the O–H stretch (broad peak around $3200\text{--}3640\text{ cm}^{-1}$) and explicitly state the number of unique proton environments.

The candidate correctly identifies CH_2 groups as triplets and associates them with chemical shifts does not specify that the triplet pattern arises from coupling with adjacent CH_2 protons ($n+1$ rule), and does not use standard structural notation (e.g., $-\text{CH}_3-\text{CH}_2-$).

The response to part (b)(iii) is incoherent and does not address the reasons for chemical shift variations.

In part (b)(iv), the student gives an incorrect structure that does not correspond to the spectrum. There are fundamental errors in both IR and NMR interpretation.

Question 7 (c)

- (a) The isomer **X** produces more energy than ethanol when burnt. To demonstrate this, the following experiment is conducted.

- (i) Write a balanced symbol equation for the complete combustion of the isomer **X**.

..... [1]

- (ii) The enthalpy change of combustion is determined. Isomer **X** is placed in a spirit burner and weighed. The burner is then lit and allowed to burn for a few minutes. The heat produced by combustion is used to warm 50 g of water in a copper calorimeter.

The burner is then re-weighed. The temperature of water is recorded before and after heating.

The results obtained are represented in **Table 1**.

Table 1

Initial mass of spirit burner and isomer X / g	205.79
Final mass of spirit burner and isomer X / g	205.60
Initial temperature of water / °C	24.5
Final temperature of water / °C	39.8

Use results from the table to calculate a value, in kJ mol^{-1} , for the enthalpy change when one mole of isomer **X** is burned.

(The specific heat capacity of water is $4.18 \text{ J K}^{-1} \text{ g}^{-1}$)

[3]

- (iii) The results from (c)(ii) are less than the standard values for the enthalpy change of combustion of butanol.

Suggest **two** reasons why the obtained results from part c (ii) are less.

1

.....

2

.....

[2]

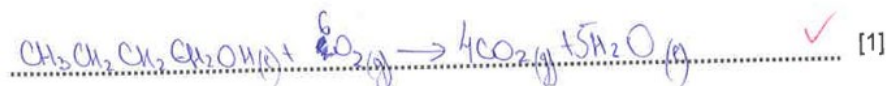
Mark Scheme

7 (c) (i)	$\text{C}_4\text{H}_9\text{OH} + 6\text{O}_2 = 4\text{CO}_2 + 5\text{H}_2\text{O}$	[1]	ecf
7 (c) (ii)	$q = 50.0 \times 4.18 \times 15.3 = 3198 \text{ (J) OR } 3.198 \text{ (kJ) OR } 3.20 \text{ (kJ) OR } 3.2 \text{ (kJ)}$ $n = 0.190 / 74.0 = 0.00257 \text{ mole}$ $\Delta H = -1244 \text{ kJ mol}^{-1}$	1 1 1 [3]	Penalise M3 ONLY if correct numerical answer but sign is incorrect. Therefore +1244 gains two marks. ecf for M3 from M1 and/or M2
7(c)(iii)	any two from • <u>heat</u> loss or not all <u>heat</u> transferred to the apparatus or <u>heat</u> absorbed by the apparatus • (specific) heat capacity of the apparatus not considered • incomplete combustion / not completely burned / reaction is not complete • reactants and / or products may not be in standard states	1 1 [2]	

Example candidate response—grade A

(c) The isomer X produces more energy than ethanol when burnt. To demonstrate this, the following experiment is conducted.

(i) Write a balanced symbol equation for the complete combustion of the isomer X.



[1]

- (ii) The enthalpy change of combustion is determined. Isomer X is placed in a spirit burner and weighed. The burner is then lit and allowed to burn for a few minutes. The heat produced by combustion is used to warm 50 g of water in a copper calorimeter.

The burner is then re-weighed. The temperature of water is recorded before and after heating.

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Use results from the table to calculate a value, in kJ mol^{-1} , for the enthalpy change when one mole of isomer X is burned.

(The specific heat capacity of water is $4.18 \text{ J K}^{-1} \text{ g}^{-1}$)

$$\Delta T = 39.8 - 24.5 = 15.3^\circ\text{C}$$

$$\text{Burned Isomer X} = 205.79 - 205.60 = 0.19 \text{ g} \quad \checkmark$$

$$q = cm\Delta T = 50 \cdot 4.18 \cdot 15.3 = 3194.4 \text{ J}$$

$$\Delta H = -\frac{q}{n} = -\frac{3194.4}{\frac{0.19}{4 \cdot 12 + 10 + 16}} = -3245420 \text{ J mol}^{-1} = -3245 \text{ kJ mol}^{-1} \quad \checkmark$$

[3]

- (iii) The results from (c)(ii) are less than the standard values for the enthalpy change of combustion of butanol.

Suggest two reasons why the obtained results from part c (ii) are less.

1. Heat was lost during experiment. $\checkmark$
2. System was not ^{not} closely insulated. $\checkmark$
3. Experiment wasn't done under standard conditions (not 298 K temperature). $\checkmark$

[2]

Examiner's comment

The candidate correctly balances the equation for the complete combustion of isomer X in excess oxygen (no CO or soot).

They use the calorimetry equation $q = mc\Delta T$ and account for the heat absorbed by the water.

The candidate correctly converts q to ΔH using the formula $\Delta H = -q/n$, where n is the number of moles of fuel.

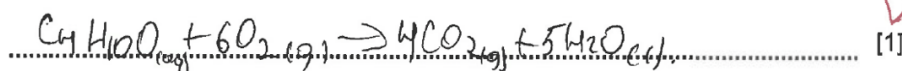
The final answer includes the correct units (kJ mol^{-1}) and the appropriate negative sign for an exothermic reaction.

In the following part of the question, the candidate states two correct reasons.

Example candidate response—grade C

- (c) The isomer X produces more energy than ethanol when burnt. To demonstrate this, the following experiment is conducted.

- (i) Write a balanced symbol equation for the complete combustion of the isomer X.



- (ii) The enthalpy change of combustion is determined. Isomer X is placed in a spirit burner and weighed. The burner is then lit and allowed to burn for a few minutes. The heat produced by combustion is used to warm 50 g of water in a copper calorimeter.

The burner is then re-weighed. The temperature of water is recorded before and after heating.

The results obtained are represented in Table 1.

Table 1

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Use results from the table to calculate a value, in kJ mol^{-1} , for the enthalpy change when one mole of isomer X is burned.

(The specific heat capacity of water is $4.18 \text{ J K}^{-1} \text{ g}^{-1}$)

$$q = mc\Delta T \quad \Delta T = 39.8 - 24.5 = 15.3$$

$$q = 4.18 \cdot 50 \cdot 15.3 = 3197.7 \text{ J}$$

$$3197.7 \text{ J} \approx 3.198 \text{ kJ}$$

$$\Delta H = \frac{q}{n}$$

$$\Delta H = \frac{3197.7}{1} = -3.198 \text{ kJ/mol}$$

[3]

- (iii) The results from (c)(ii) are less than the standard values for the enthalpy change of combustion of butanol.

Suggest two reasons why the obtained results from part c (ii) are less.

1 Heat loss would take place during experiment.

Not ~~an~~ appropriate isolation for heat was used.

2 Used apparatus ~~was~~ had some impurities.

which led to the decrease of mass of butanol.

[2]

Examiner's comment

The candidate calculates the temperature change (ΔT) correctly and applies the formula $q = mc\Delta T$ to find the energy released in joules.

However, the conversion to kilojoules is incorrect, and the enthalpy change is not calculated using the correct number of moles.

The first suggested reason — heat loss to the surroundings — is valid and clearly explained. The second reason is poorly expressed and lacks scientific accuracy.

Example candidate response—grade E

(c) The isomer X produces more energy than ethanol when burnt. To demonstrate this, the following experiment is conducted.

(i) Write a balanced symbol equation for the complete combustion of the isomer X.



(ii) The enthalpy change of combustion is determined. Isomer X is placed in a spirit burner and weighed. The burner is then lit and allowed to burn for a few minutes. The heat produced by combustion is used to warm 50 g of water in a copper calorimeter.

The burner is then re-weighed. The temperature of water is recorded before and after heating.

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Use results from the table to calculate a value, in kJ mol^{-1} , for the enthalpy change when one mole of isomer X is burned.
(The specific heat capacity of water is $4.18 \text{ J K}^{-1} \text{ g}^{-1}$)

$\Delta T = 15.3^\circ\text{C} = 288.3\text{K}$
 $\Delta m = -0.19$
 $c = 4.18$
 $\Delta H = \Delta m c \Delta T = -0.19 \cdot 4.18 \cdot 288.35$
 $= -229 \text{ J}$ [3]

(iii) The results from (c)(ii) are less than the standard values for the enthalpy change of combustion of butanol.

Suggest two reasons why the obtained results from part c (ii) are less.

- 1 because ^{heat} temperature is gone to a system
- 2 mass burned in to light energy it is used up to warm water

Examiner's comment

The candidate uses an incorrect formula for oxygen and makes no attempt to balance the equation correctly, therefore, no marks are awarded. They misapply the equation $q = mc\Delta T$, using it as equivalent to ΔH , without establishing a proper link between q and ΔH . Key calculation steps are completely omitted. In the evaluation section, the candidate fails to consider possible experimental errors and instead provides vague, inaccurate general statements.

Question 7 (d)

(c) Bond energies can also be used to calculate the standard enthalpy of combustion of a compound.

(i) Use the *Data Booklet* to calculate a value for the standard enthalpy of combustion of isomer **X**. Show your working.

[3]

(ii) The standard enthalpy of combustion of isomer **X** is $-2670 \text{ kJ mol}^{-1}$. Explain why this value differs from the value obtained using bond energies.

.....

.....

.....

..... [2]

Mark Scheme

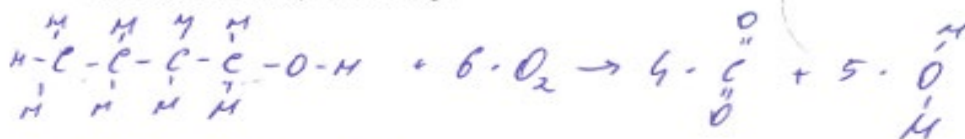
7(d)(i)	M1 $\Delta H = 8536$ M2 $\Delta H = 10520$ $\Delta H = [8536] - [10520] = -1984 \text{ (kJ mol}^{-1}\text{)}$	1 1 1 [3]	ecf
7(d)(ii)	Bond energies calculated by using enthalpy changes of gaseous compound/standard conditions Enthalpy change for combustion reaction is obtained by using liquid compound /the bond energy in the data booklet is an average value / bond energy value for C=O in data booklet is not the value for the C=O bond in carbon dioxide.	1 1 [2]	OWTTE

Example candidate response — grade A

(d) Bond energies can also be used to calculate the standard enthalpy of combustion of a compound.

For
Examiner's
Use

(i) Use the *Data Booklet* to calculate a value for the standard enthalpy of combustion of isomer X. Show your working.



$$\begin{aligned} E_{\text{before}} &= 3 \cdot (\text{C}-\text{C}) + 9 \cdot (\text{C}-\text{H}) + \text{C}-\text{O} + \text{O}-\text{H} + 6 \cdot (\text{O}=\text{O}) = \\ &= 3 \cdot 350 + 9 \cdot 410 + 360 + 460 + 6 \cdot 496 = \\ &= \cancel{8886} \quad 8536 \end{aligned}$$

$$\begin{aligned} E_{\text{after}} &= 4 \cdot 2 \cdot (\text{C}=\text{O}) + 5 \cdot 2 \cdot (\text{O}-\text{H}) = 8 \cdot 740 + 10 \cdot 460 = \\ &= 10520 \end{aligned}$$

$$\Delta H = E_{\text{after}} - E_{\text{before}} = 10520 - 8536 = 1984 \text{ kJ mol}^{-1}$$

[3]

(ii) The standard enthalpy of combustion of isomer X is $-2670 \text{ kJ mol}^{-1}$. Explain why this value differs from the value obtained using bond energies.

mean bond energy is used for calculation,
which is calculated for a range of compounds,
not 1 specifically.

[2]

[Total: 16]

Examiner's comment

The candidate uses bond energies and performs the calculation for the standard enthalpy of combustion using two bond energy values, earning 3 marks.

For the explanation of the difference between the given enthalpy of combustion and the value obtained using bond energies, the candidate provides only one reason and does not account for the states of the compounds. Therefore, only partial marks are awarded for this part.

Example candidate response — grade C

(d) Bond energies can also be used to calculate the standard enthalpy of combustion of a compound.

(i) Use the *Data Booklet* to calculate a value for the standard enthalpy of combustion of isomer X. Show your working.

For
Examiner's
Use



C-C-C-C : 3 × 350

C-H : 9 × 410

O-H : 460

C-O : 360

O=O : 6 × 496

C=O : 2 × 4 × 740

H-O : 5 × 2 × 460

formed = 10520

broken
= 8536

~~$\Delta H_c = \text{formed} - \text{broken}$~~

$$\Delta H_c = \text{broken} - \text{formed} = 8536 - 10520 = -1984$$

$$\Delta H_c = -1984 \frac{\text{kJ}}{\text{mol}}$$

[3]

(ii) The standard enthalpy of combustion of isomer X is $-2670 \text{ kJ mol}^{-1}$. Explain why this value differs from the value obtained using bond energies.

isomer X is more branched than straight chain butanol. As a result, it ~~is~~ isomer has less surface area to react with, resulting in less combustion.

[2]

Examiner's comment

The candidate uses bond energies and performs the calculation for the standard enthalpy of combustion using two bond energy values, earning 3 marks.

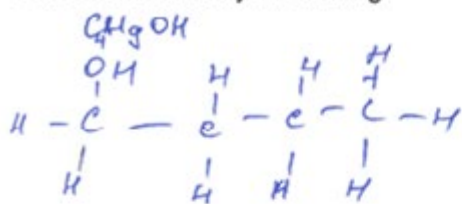
However, in the following step, the student incorrectly explains the difference between the two values, giving an inaccurate explanation related to the structure of the compound and its surface area.

Example candidate response—grade E

(d) Bond energies can also be used to calculate the standard enthalpy of combustion of a compound.

For
Examiner's
Use

(i) Use the Data Booklet to calculate a value for the standard enthalpy of combustion of isomer X. Show your working.



$$\begin{aligned}
 \Delta H(4 \text{ CO}_2) &= 5920 \\
 \Delta H(5 \text{ H}_2\text{O}) &= 4600
 \end{aligned}$$

$$350 \cdot 4 + 410 \cdot 9 + 360 + 460 =$$

$$= 5910 \text{ kJ mol}^{-1}$$

$$\Delta H(6 \text{ O}_2) = 2946$$

$$5910 + 2946 - (5920 + 4600) =$$

$$= -1634 \text{ kJ mol}^{-1}$$

X✓X [3]

(ii) The standard enthalpy of combustion of isomer X is $-2670 \text{ kJ mol}^{-1}$. Explain why this value differs from the value obtained using bond energies.

Because the small part of heat goes to the environment

XX [2]

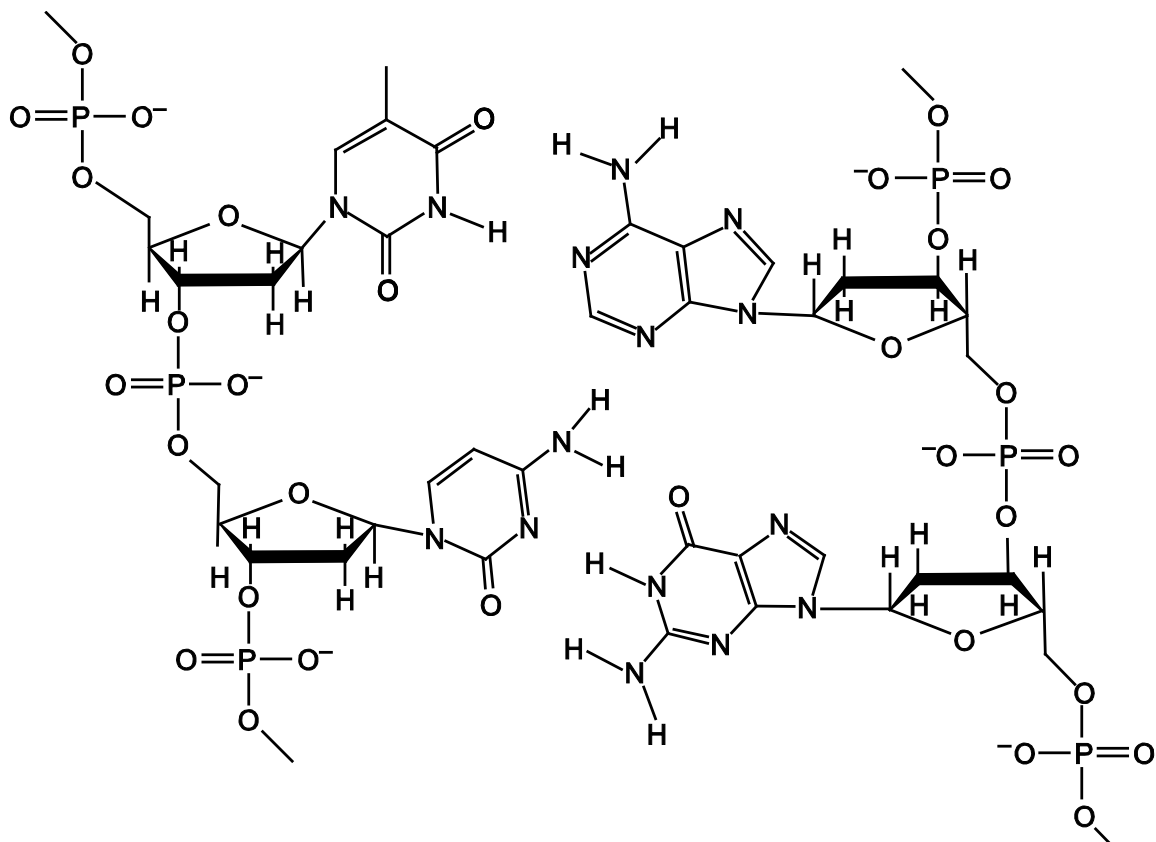
Examiner's comment

The candidate uses bond energies but miscalculates the total bond energy for the reactants. However, the calculation for the products is correct. As a result, only 1 out of 3 marks is awarded for the calculation.

The explanation for the difference between the given enthalpy of combustion and the calculated value is incorrect, therefore, no marks are awarded for this part.

Question 8 (a)

A single strand of DNA is a polymer – it is made up of four different monomers, nucleotides bonded together. These monomers are called nitrogenous bases.



(a) DNA exists as two strands of nucleotides in the form of a double helix with hydrogen bonding between the strands.

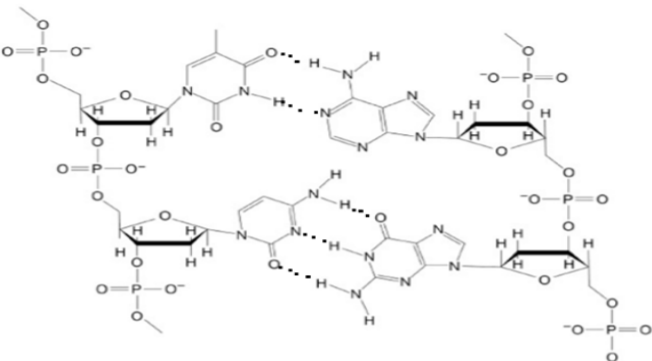
(i) Use these structures of nitrogenous bases to show hydrogen bonds between base pairs.

[2]

(ii) Name the correct complementary base pairing for DNA.

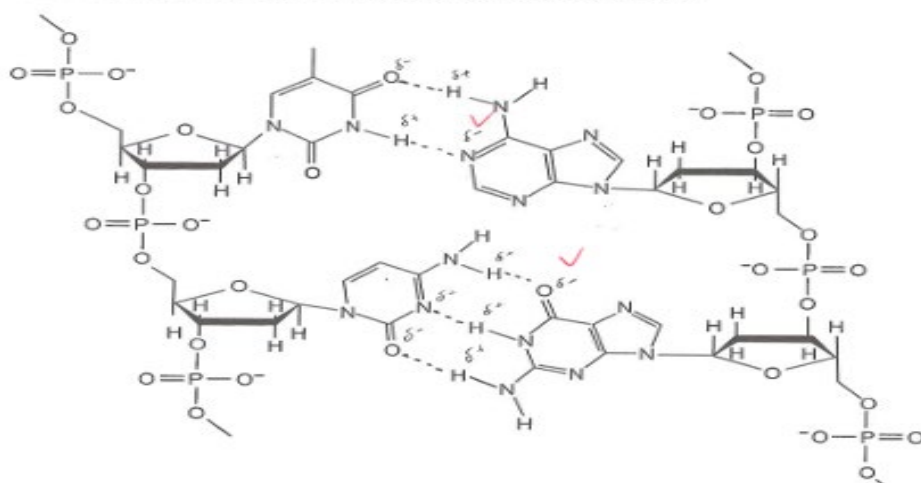
.....
..... [1]

Mark Scheme

8(a)(i)	 2 hydrogen bonds shown between adenine and thymine 3 hydrogen bonds shown between guanine and cytosine	1 1 [2]	
8(a)(ii)	Thymine and adenine AND cytosine and guanine.	[1]	

Example candidate response—grade A

- 8 A single strand of DNA is a polymer – it is made up of four different monomers, nucleotides bonded together. These monomers are called nitrogenous bases.



- (a) DNA exists as two strands of nucleotides in the form of a double helix with hydrogen bonding between the strands.

- (i) Use these structures of nitrogenous bases to show hydrogen bonds between base pairs.

[2]

- (ii) Name the correct complementary base pairing for DNA.

thymine - adenine (A = T), guanine - cytosine (G = C)

[1]

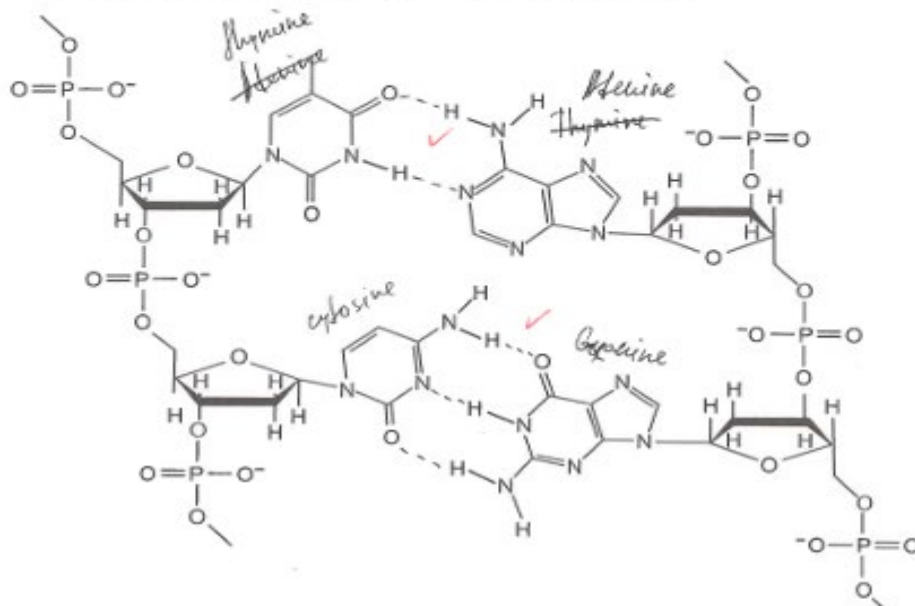
Examiner's comment

The candidate correctly draws three hydrogen bonds between **G≡C**, and two hydrogen bonds between **A=T**, using dashed lines with proper orientation. Full marks are awarded (2 marks). The candidate also clearly states that guanine (G) pairs with cytosine (C) and adenine (A) pairs with thymine (T).

Example candidate response—grade C

- 8 A single strand of DNA is a polymer – it is made up of four different monomers, nucleotides bonded together. These monomers are called nitrogenous bases.

For
Examiner's
Use



- (a) DNA exists as two strands of nucleotides in the form of a double helix with hydrogen bonding between the strands.

- (i) Use these structures of nitrogenous bases to show hydrogen bonds between base pairs.

[2]

2

- (ii) Name the correct complementary base pairing for DNA.

Adenine = Thymine X

[1]

0

Examiner's comment

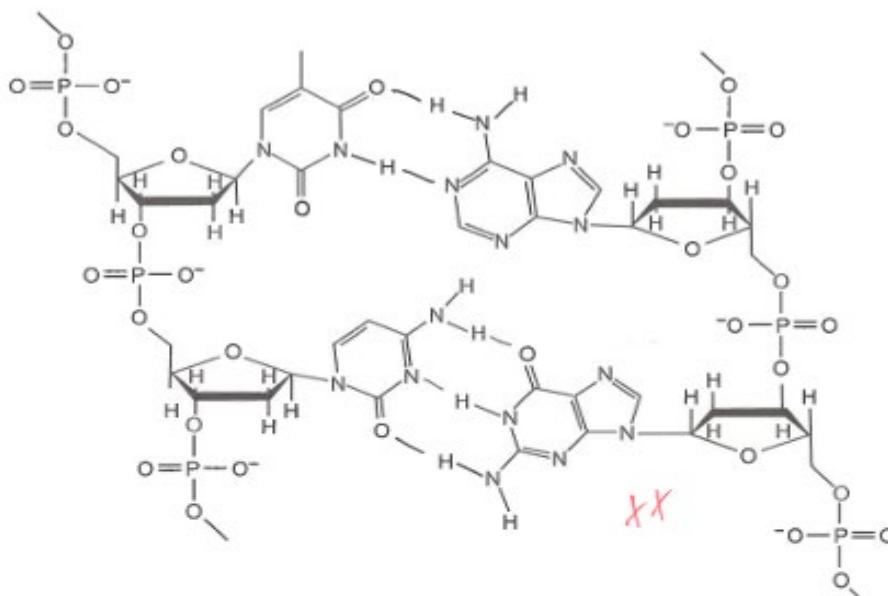
The candidate correctly draws three hydrogen bonds between **G≡C** and two hydrogen bonds between **A=T**, using dashed lines with proper orientation.

The candidate names the A–T base pair but omits G–C, therefore, no mark is awarded for the base pairing identification.

Example candidate response—grade E

- 8 A single strand of DNA is a polymer – it is made up of four different monomers, nucleotides bonded together. These monomers are called nitrogenous bases.

For
Examiner's
Use



- (a) DNA exists as two strands of nucleotides in the form of a double helix with hydrogen bonding between the strands.

- (i) Use these structures of nitrogenous bases to show hydrogen bonds between base pairs.

$C \equiv G$ ~~double~~ double and triple
 $A = T$ bonds are formed

[2]

- (ii) Name the correct complementary base pairing for DNA.

Adenine - Thymine

Cytosine - Guanine

[1]

Examiner's comment

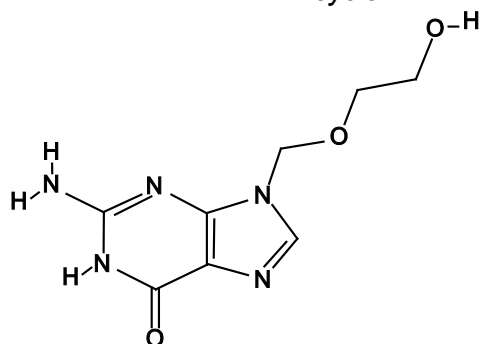
The candidate correctly identifies the base-pairing regions but incorrectly represents the hydrogen bonds using solid lines instead of dashed lines. Therefore, no marks are awarded for the diagram.

However, they clearly state that guanine (G) pairs with cytosine (C), and adenine (A) pairs with thymine (T).

Question 8 (b)

- (b) Acyclovir tricks the viral enzymes into using it as building blocks of DNA and thus prevents the virus from multiplying.

The structure of *Acyclovir*



- (i) Indicate the amide bond in the structure of the acyclovir molecule.

[1]

- (ii) Suggest why it is better to take medicine by injection rather than by mouth.

..... [1]

- (iii) Describe what side effects may occur when taking medications and how to eliminate them.

..... [1]

Mark Scheme

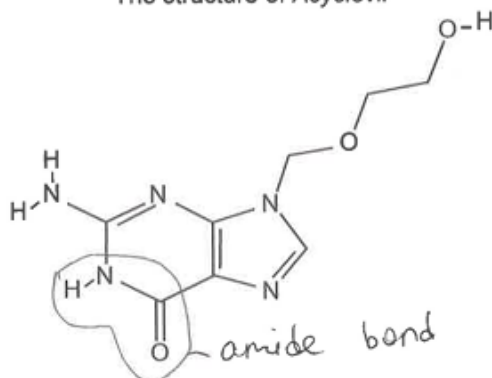
8(b)(i)		[1]	
8(b)(ii)	- the acidity of the stomach before it can be absorbed into the blood/ - damage to the drug molecules at very low pH - prevent hydrolysis/less amount drugs/more active substance	[1]	AVP
8(b)(iii)	Headache/ nausea/ vomiting/ abdominal pain and Stop taking the drug and see a doctor	[1]	

Example candidate response—grade A

- (b) Acyclovir tricks the viral enzymes into using it as building blocks of DNA and thus prevents the virus from multiplying.

For
Examiner's
Use

The structure of Acyclovir



- (i) Indicate the amide bond in the structure of the acyclovir molecule.

[1]

- (ii) Suggest why it is better to take medicine by injection rather than by mouth.

faster rate of absorption, medicines injected by mouth can be affected by enzymes before reaching stomach, F.E by insulin [1]

- (iii) Describe what side effects may occur when taking medications and how to eliminate them.

allergic reactions to medications to eliminate them always read instructions from what is this medication is consist of and go to the doctor's advice [1]

Examiner's comment

The candidate correctly indicates the amide bond in the structure of the Acyclovir molecule. 1 mark is awarded.

The explanation in part (b)(ii) is incomplete but demonstrates a general understanding that injection bypasses the digestive system, leading to faster and more complete absorption. They also note that oral medicines can be broken down by digestive enzymes. 1 mark is awarded.

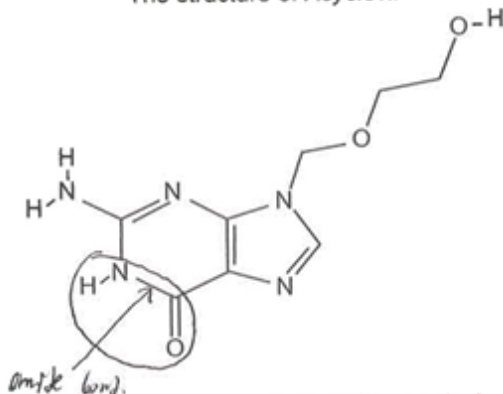
In part (b)(iii), the candidate mentions allergic reactions and refers to elimination methods such as following the instructions provided with the medication and consulting a doctor. 1 mark is awarded.

Example candidate response—grade C

- (b) Acyclovir tricks the viral enzymes into using it as building blocks of DNA and thus prevents the virus from multiplying.

For
Examiner's
Use

The structure of Acyclovir



- (i) Indicate the amide bond in the structure of the acyclovir molecule.

[1]

- (ii) Suggest why it is better to take medicine by injection rather than by mouth.

Because it works faster, absorb faster.
Drug can not be hydrolysed in stomach.

[1]

- (iii) Describe what side effects may occur when taking medications and how to eliminate them.

Vomiting → use injection, Allergic reaction →
Use other drugs.

[1]

Examiner's comment

The candidate correctly indicates the amide bond in the structure of the Acyclovir molecule. 1 mark is awarded.

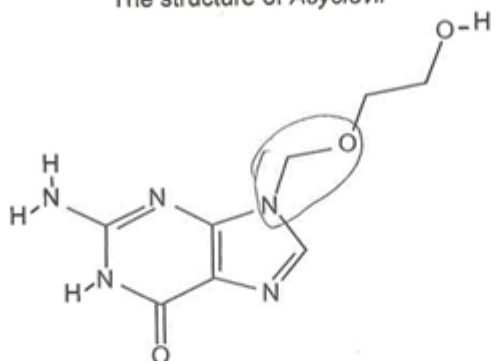
However, in part (b)(iii), the candidate's answer does not address any valid method for eliminating side effects, therefore, no mark is awarded for this part.

Example candidate response—grade E

- (b) Acyclovir tricks the viral enzymes into using it as building blocks of DNA and thus prevents the virus from multiplying.

For
Examiner's
Use

The structure of Acyclovir



- (i) Indicate the amide bond in the structure of the acyclovir molecule.

X

[1]

- (ii) Suggest why it is better to take medicine by injection rather than by mouth.

because the medicine can be decomposed by mouth
ferments before it reaches the organism. [1]

- (iii) Describe what side effects may occur when taking medications and how to eliminate them.

allergic reactions, consulting from doctor to
know which medications you can
use without side effects. [1]

Examiner's comment

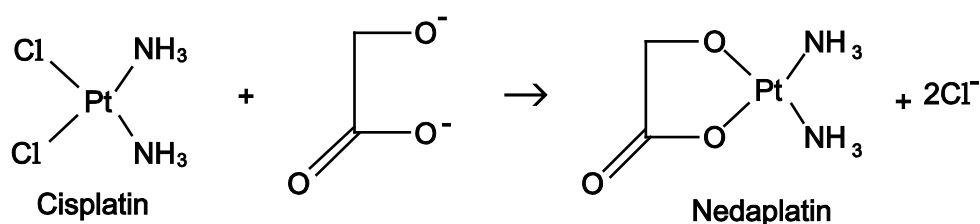
The candidate incorrectly identifies the amide bond in the structure of the Acyclovir molecule. The explanation in part (b)(ii) misses critical factors, such as the effect of stomach acid and first-pass metabolism.

In part (b)(iii), the response lacks specific examples or practical steps for managing side effects.

Question 8 (c)

(c) *Nedaplatin*, a new anticancer drug with less side effects, has been developed.

It can be synthesised from *cis*-platin by the following reaction.



(i) Explain why the reaction occurs in terms of the chelate effect.

.....

.....

.....

..... [3]

(ii) In recent years, scientists have developed a variety of different drug delivery systems with less side effects. They found out that gold nano particles can be effectively used in cancer treatment.

Suggest how gold nano-cages selectively deliver cancer-fighting drugs to the sites of tumours in the body.

.....

.....

.....

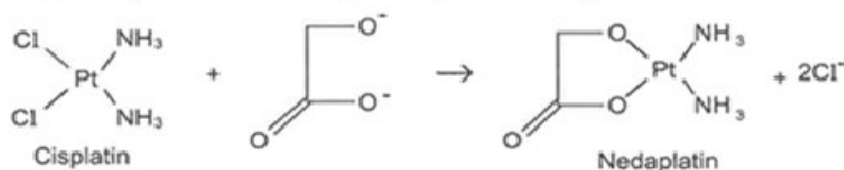
..... [2]

Mark Scheme

8 (c) (i)	Increase in number of molecules/ 2 moles to 3 <u>moles</u> so entropy change/ ΔS is positive similar type of bonds broken and formed so enthalpy change ΔH is approximately zero / negligible / bidentate ligand and multidentate <u>K_{stab}</u> is higher/ stable product/ΔG negative	1 1 1 [3]	OWTTE
8(c)(ii)	<u>Tumours</u> have thin blood vessels with holes <u>large</u> enough/PEG <u>So</u> nanoparticle can <u>pass</u> into a <u>tumour</u>/heat released/change shape and opening pores	1 1 [2]	ORA

Example candidate response—grade A

- (c) Nedaplatin, a new anticancer drug with less side effects, has been developed. It can be synthesised from cis-platin by the following reaction.



- (i) Explain why the reaction occurs in terms of the chelate effect.

Cl^- is monodentate, $\text{C}_2\text{O}_4^{2-}$ is bidentate ligand. According to chelate effect, the more co-ordinate bond can be formed (bidentate for example) the more the molecule is protected. Then entropy is more because more particles are produced, so ΔG is negative, it occurs.

(ii) In recent years, scientists have developed a variety of different drug delivery systems with less side effects. They found out that gold nano particles can be effectively used in cancer treatment.

Suggest how gold nano-cages selectively deliver cancer-fighting drugs to the sites of tumours in the body.

They are very small $10^{-7} - 10^{-8}$, so it adjacent to DNA and easily, quickly identify the part that has been undergone mutation. And gold nano-cages not absorbed by any other side of body and goes directly to the sites of tumors.

Examiner's comment

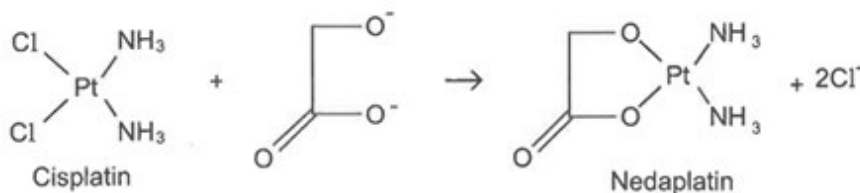
The candidate correctly states that oxalate is a bidentate ligand, mentions the increase in entropy, and identifies that Gibbs free energy is negative. 3 marks are awarded.

In part (c)(ii), they refer to both the size of the nanocage and action. 2 marks are awarded.

Example candidate response—grade C

- (c) *Nedaplatin*, a new anticancer drug with less side effects, has been developed. It can be synthesised from *cis*-platin by the following reaction.

For
Examiner's
Use



- (i) Explain why the reaction occurs in terms of the chelate effect.

Because bidentate ligands ✓ has two pairing electrons. And they're much more stable ✓ compared to monodentate ligands. They form very stable complex ion. [3] 2

- (ii) In recent years, scientists have developed a variety of different drug delivery systems with less side effects. They found out that gold nano particles can be effectively used in cancer treatment.

Suggest how gold nano-cages selectively deliver cancer-fighting drugs to the sites of tumours in the body.

They have very good area to react. As they're tiny. ✓ Drug delivered faster and reaction happens faster. [2] 1

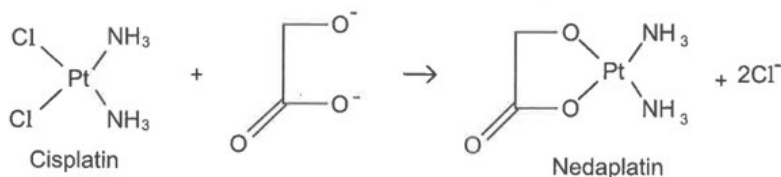
Examiner's comment

The candidate recognises the increased stability but incorrectly refers to the complex as an "ion". 2 marks are awarded.

In part (c)(ii), the candidate correctly states the size of the nanoparticles but does not provide any explanation about their action. 1 mark is awarded.

Example candidate response—grade E

- (c) Nedaplatin, a new anticancer drug with less side effects, has been developed. It can be synthesised from cis-platin by the following reaction.



- (i) Explain why the reaction occurs in terms of the chelate effect.

Cisplatin is non-polar and it reacts with polar substance through electrophilic addition mechanism creating Nedaplatin and giving 2Cl⁻ atoms as residue. +++ [3]

- (ii) In recent years, scientists have developed a variety of different drug delivery systems with less side effects. They found out that gold nano particles can be effectively used in cancer treatment.

Suggest how gold nano-cages selectively deliver cancer-fighting drugs to the sites of tumours in the body.

Drugs can be injected into the gold nano-cages and cages will protect drugs from spilling because of saliva or acid in the mouth and belly ensuring that the drug will be delivered to the cells. XX [2]

For
Examiner's
Use

0

0

Examiner's comment

The candidate provides no valid reasoning and appears to confuse the reaction mechanism with thermodynamic concepts such as stability. No marks are awarded for this part. The explanation of the action of nanoparticles is incorrect. No marks are awarded for this part either.

Question 8 (d) (i), (ii)

(d) *Cis*-platin has a geometrical isomer.

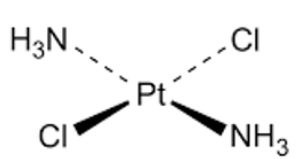
(i) Draw 3D structure of its isomer – *trans*-platin.

[1]

(ii) Explain why *trans*-platin is not used as anticancer drug.

.....
..... [1]

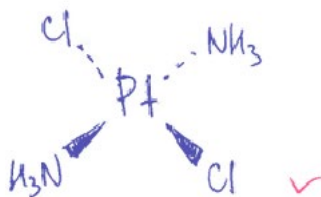
Mark Scheme

8 (d) (i)		[1]	Accept planar structure
8(d) (ii)	Cl ⁻ ligands located in opposite direction so can not form crosslink with DNA	[1]	OWTTE

Example candidate response—grade A

(d) *Cis-platin* has a geometrical isomer.

(i) Draw 3D structure of its isomer – *trans-platin*.



[1]

1

(ii) Explain why *trans-platin* is not used as anticancer drug.

because its Cl ligands are on the opposite side so it cannot bind to guanine in DNA to postpone replication and to stop tumour growth. [1]

1

Examiner's comment

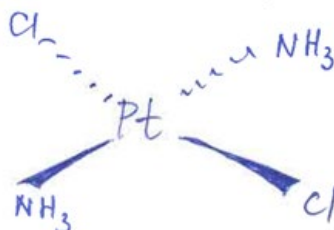
The candidate correctly draws the square planar structure of cisplatin, showing Cl and NH₃ ligands in the cis positions with appropriate 90° bond angles.

The explanation of why trans-platin is ineffective as an anticancer drug is nearly correct, noting the opposite positions of the Cl ligands.

Example candidate response — grade C

(d) *Cis-platin* has a geometrical isomer.

(i) Draw 3D structure of its isomer – *trans-platin*.



[1]

✓

1

(ii) Explain why *trans-platin* is not used as anticancer drug.

Not so effective, the chemical property is different. might be toxic. [1]

X

0

Examiner's comment

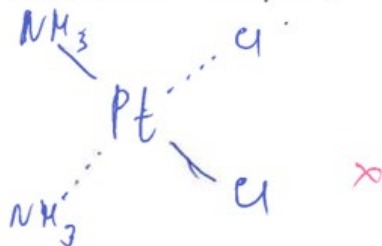
The candidate correctly shows the square planar structure of the molecule.

However, the explanation includes incorrect statements about the properties of the molecule, therefore, no mark is awarded for the reasoning.

Example candidate response — grade E

(d) *Cis*-platin has a geometrical isomer.

(i) Draw 3D structure of its isomer – *trans*-platin.



[1]

(ii) Explain why *trans*-platin is not used as anticancer drug.

Because it ^{has} different chemical properties.

[1]

Examiner's comment

The candidate makes fundamental errors in the structure of *trans*-platin. Additionally, the explanation includes incorrect statements about the properties of the molecule. No marks are awarded.